Enhancing Efficiency and Robustness for Switchback Experiments: A Model-assisted Design and Analysis

Tu Ni
Digital, Data, Design Institute, Harvard Business School, Boston, MA 02163, tni@hbs.edu

Iavor Bojinov
Technology and Operations Management Unit, Harvard Business School, Boston, MA 02163, ibojinov@hbs.edu

Switchback experiments have emerged as a powerful alternative when traditional A/B tests are constrained by limits on user-level randomization or by interference among experimental units. However, in the presence of carryover effects, existing switchback designs often mitigate bias by randomly discarding data—resulting in reduced statistical power and an unpredictable yield of valid observations. We propose a model-assisted design that harnesses two pervasive temporal features—seasonal and serial autocorrelations—identified via flexible time-series modeling to inform treatment assignments. Our method employs an adaptive block randomization scheme that evenly allocates observations across seasonal cycles, guaranteeing treatment–control balance and dampening sensitivity to peak periods. A temporal-coupling refinement further leverages recent treatment history to balance assignments in short time windows. We analyze the bias — arising from the use of short-term data to estimate long-term causal effects — and variance of our designs and develop corresponding inferential procedures. In realistic ride-sharing simulations, our design achieves up to a 30 % reduction in RMSE relative to current standards. By marrying practical considerations with statistical techniques, this work advances experimental design in the presence of interference and rich temporal dynamics.

Key words: switchback experiments; causal inference; time series; carryover effects

1. Introduction

Companies are increasingly focusing on developing advanced data science and artificial intelligence (AI) solutions to optimize the performance of different business metrics. These solutions could be complex, comprising multiple parameters and features that require thorough validation through experimentation, which has become a standard practice for evaluating the impact of innovations and changes to products and services before wide-scale release in modern business operations(Thomke 2003, Kohavi and Thomke 2017, Cui et al. 2018, Gupta et al. 2019, Bojinov and Gupta 2022, Larsen et al. 2023, Bojinov et al. 2025). The most prominent experimental design is the standard A/B test in which experimental subjects (e.g., users) are randomly assigned to the change (often called the treatment), while the rest receive the status quo (often called the control); the impact is then

estimated by comparing the two groups across various business metrics or outcomes associated with the new solution.

Unfortunately, traditional A/B testing methods are not feasible in some markets or environments due to constraints in user randomization. For example, consumer packaged goods (CPG) companies that sell products on third-party platforms regularly test various strategies for pricing, auction algorithms, and assortments. Unfortunately, these CPG companies do not directly interact with users and are unable to randomize them to alternative treatments, making it impossible to run a standard A/B test. Moreover, even when it is possible to run A/B tests, these experiments often suffer from population interference, the process by which the actions or experiences of individuals assigned to the treatment group inadvertently affect the outcomes of the people in the control group (for example, Airbnb (Li et al. 2022), DoorDash (Tang et al. 2020), LinkedIn (Saveski et al. 2017), and Lyft (Chamandy 2016)). As a result, the non-interference part of the standard stable-unit treatment value assumption (SUTVA) (Rubin 1980, Holland 1986) is violated, complicating the estimation of true causal effects.

Given these limitations of A/B tests, switchback experiments have emerged as a promising alternative (Bojinov and Shephard 2019, Bojinov et al. 2022, Glynn et al. 2020, Hu and Wager 2022, Xiong et al. 2022, Chen and Simchi-Levi 2023, Ni et al. 2023, Jia et al. 2024, Wen et al. 2024, Sun et al. 2024, Missault et al. 2025, Ni 2025). The basic idea to mitigate population interference is to group individuals into clusters so that there is no interference across clusters; the experimenter then changes the treatment assignments over time (Menchetti and Bojinov 2022, Ham et al. 2022). Switchback experiments represent the most extreme case where there is a single cluster — or unit. In such an experiment, the unit (e.g., all drivers on a ride-sharing platform in the Boston metropolitan area) is randomly assigned to either treatment or control (by convention, we denote treatment by 1 and control by 0), its outcome is recorded, and the process is repeated for T time periods creating a series of treatment assignments (e.g., "1001101110...") and corresponding outcomes. Inference then focuses on the average responses over time with the goal of determining if the treatment is better than the control. Although the grouping procedure removes the concerns posed by population interference, it comes at the cost of a drastic reduction in the effective sample size: from thousands and sometimes millions of individuals to a few hundred time periods. For example, if the duration of a period is one hour, then a two-week experimental horizon maps to $T = 24 \times 14 = 336$ periods, which may not be sufficient for detecting the relatively small effects that are common in well optimized businesses.

The power of switchback experiments further deteriorates in the presence of carryover effects, where prior treatment assignments influence current outcomes (Cox 1958). An expanding literature addresses the design and analysis of switchback experiments under such effects (Bojinov et al. 2022,

Hu and Wager 2022, Jia et al. 2024, Ni et al. 2023, Chen and Simchi-Levi 2023, Xiong et al. 2024, Missault et al. 2025). Despite differences in theoretical frameworks, most approaches share a similar estimation principle: they leverage appropriately defined local outcomes (e.g., outcomes under two consecutive treatments) to approximate global outcomes (e.g., under the full sequence of treatments). This typically requires discarding a subset of observations after each switching based on the local definition — if the local radius is too small, the resulting discrepancy between local (short-term) and global (long-term) outcomes can introduce substantial bias, whereas if the radius is too large, discarding too many observations diminishes the effective sample size and increases the variance. Consequently, the central challenge is to develop methods for strategically discarding data in order to achieve a favorable balance between bias and variance. Notably, Xiong et al. (2024) propose a practical, data-driven approach to this problem, introducing an empirical Bayes design. Beyond this, however, practical considerations and concerns are hardly studied in the literature.

1.1. Practical Concerns and Challenges

We summarize two main concerns and challenges that are critical to the successful implementation of switchback experiments. The first relates more to the design of experiments, while the second pertains more to the analysis of experiments.

- **Balancing challenges with sample-size volatility.** Unlike the classical experiments with the completely randomized design where the number of treatment and control observations is fixed and equal (i,e., half and half), the process of discarding observations to mitigate carryover effects introduces randomness into the effective sample size, potentially yielding unstable results. For instance, while an assignment sequence such as "111110000" might retain a majority of observations, it remains vulnerable to systematic shifts before and after the switch, resulting in substantial variability. Conversely, an assignment sequence like "1010011010" is less prone to systematic shifts but will yield fewer valid observations, thereby severely diminishing the experiment's statistical power. Moreover, balancing is a powerful principle in experimental design where similar units are assigned to treatment and control groups. For example, peak hours – say, 7–8 p.m., when sales, rides or clicks spike – should be evenly represented; otherwise, treating most peaks and controlling most troughs can severely reduce the estimation precision. However, achieving such balance under carryover is non-trivial. Randomness in the effective sample size and serial correlation in time periods make realized balance a moving target: even if the allocated assignments are perfectly stratified, the effective samples can still be imbalanced, inflating the variance of the treatment effect estimate.
- **Limitations of model-based and model-free approaches.** There are two main streams of methodologies in analyzing the switchback experiments. On the one hand, model-based

approaches have been employed to better capture system dynamics and enhance statistical power by imposing a specific parametric structure on the data-generating process (Sun et al. 2024). Although these methods can lead to more efficient design and analysis when the underlying assumptions hold, practically in complex operational contexts where factors, such as customer behavior and competitors' actions, are unlikely to hold. When the assumptions are violated, the results will likely be biased and produce invalid inferences, ultimately undermining the purpose of conducting an experiment: to obtain reliable information that can be used to augment decision-making. On the other, model-free approaches have been developed to alleviate the burden of justifying parametric assumptions, offering non-parametric methods for inferring the treatment effect (Bojinov et al. 2022, Ni et al. 2023). However, these methods typically rely on other structural assumptions about the outcomes, such as the limited carryover effect that requires the treatment's impact dissipates completely after a certain period. Moreover, model-free methods may exhibit lower efficiency compared to model-based approaches, potentially resulting in wider confidence intervals and less precise estimates.

1.2. Main Results and Contributions

This paper contributes to both the theoretical development of experimental design and the practical implementation of switchback experiments by leveraging a new model-assisted framework, addressing the concerns above.

Consider a ride-hailing platform experimenting with its matching or pricing algorithms. Such platforms typically maintain extensive historical data on past trips. For example, the number of trips per hour can be easily aggregated, and such descriptive statistics often correlate strongly with key business metrics such as revenue, wait time, and travel time (See Figure 1 for the NYC taxi data (Taxi and Commission 2025)). To inform our design, we take a model-assisted view: we first use the historical data to fit a time series model, uncovering empirically observed patterns in the system. In particular, we can perform an autocorrelation analysis for the time series data, as shown in Figure 2. While we do not commit to a specific functional form, fitting general time-series models to such data reliably reveals two fundamental features: seasonality component, which captures recurring patterns (e.g., every 24 hours in Figure 2), and serial autocorrelation, which reflects temporal dependencies in local outcomes (e.g., positive correlation with small lags in Figure 2). These properties emerge robustly under flexible modeling assumptions and guide our experimental design.

Our design extends the balancing principle to switchback experiments with carryover effects. We begin by identifying the natural seasonality cycle from the data—for instance, every 24 hours as the daily pattern. We then introduce an adaptive randomization procedure governed by a key parameter ρ, which controls the switching frequency. For every hour of the day, among the total $T/24$ time

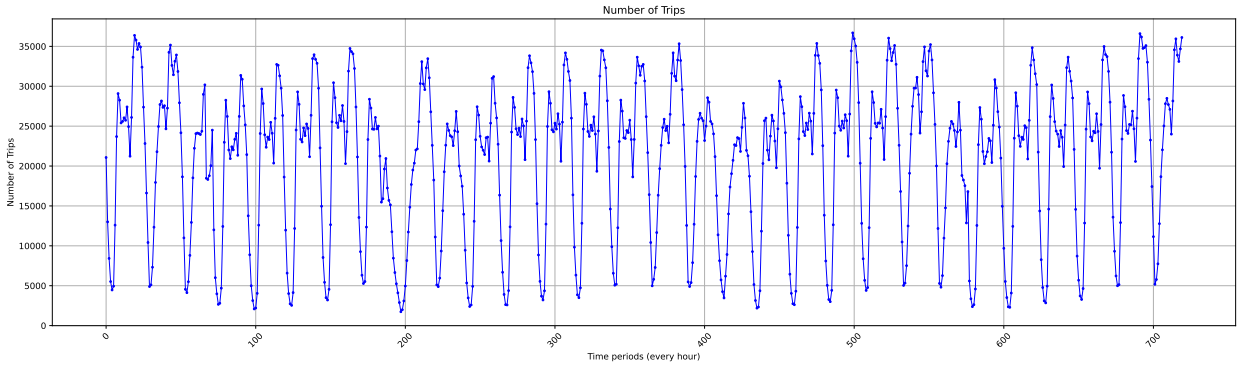


Figure 1 **Number of trips from NYC taxi data in 2013-02**

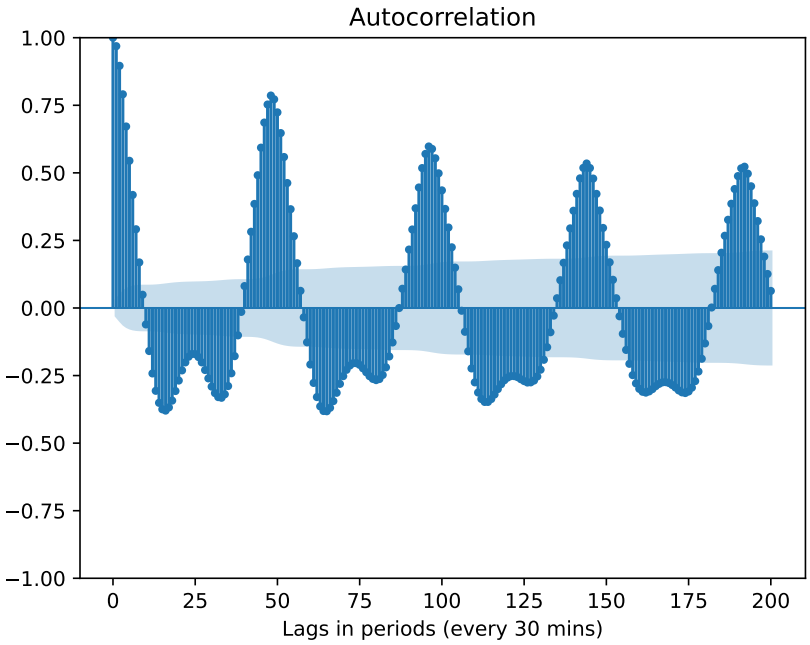


Figure 2 **The autocorrelation of the time series in lags**

periods, we collect $\rho T/24$ effective samples that are split equally into treatment and control groups completely at random. This guarantees uniform coverage across the cycle while maintaining balance between treatment and control groups, reducing sensitivity to peak periods and improving estimation efficiency.

The choice of the parameter ρ reflects the tradeoff between seasonal and serial autocorrelations. A large ρ results in more data for each hour of the day but fewer switches within the day. This is desirable when the seasonality component dominates and serial autocorrelation is weak; a small ρ increases switching within the day but gathers less per-hour data. This is preferred when serial autocorrelation is strong and the seasonality component is less pronounced. Importantly, the design

guarantees that for any realized assignment path over T periods, ρT number of observations are effective, while the remaining $(1-\rho)T$ observations are discarded or burned in the analysis. Since the number of effective samples collected from an experiment is predetermined, decision-makers can ensure the experiment will yield high-quality results before starting. This allows for more deliberate and reliable experimentation, while also facilitating power analysis.

While we do leverage historical data to exploit a fundamental property common to most temporal or dynamic systems, our estimation and inference methods are fully non-parametric. In other words, the causal conclusion does not rely on any predetermined functional form or specific outcome model. Importantly, our analytical results remain valid even if these model-assisted characteristics are not accurate, which ensures robust and practical experimentation.

We conduct a comprehensive analysis of the bias and variance associated with the new design, highlighting the influence of the design parameter ρ. Intuitively, a large ρ forces longer unchanged assignments such as "11111.." or "00000..". Those sequences give the system more time to "settle" into a pure treatment (or control) state, so the measured outcome is closer to what we would observe in a world that was always treated (or always control). As a result, the impact of the carryover effect shrinks and the bias falls as ρ rises. However, the variance does not necessarily exhibit monotonicity — either small or large ρ can minimize variance, depending on the similarity of potential outcomes across periods. Building on this new design, we derive a central limit theorem (CLT) and develop a variance estimator to facilitate statistical inference.

Furthermore, we refine our new design by generating the assignments more adaptively based on the assignments of past periods; for instance, by increasing the probability of assigning control after two consecutive treatments. This mechanism temporally couples treatment and control observations within reasonably short time windows. The cost of such refinement is the introduction of an additional design parameter ρ^+, whose impact on bias and variance is characterized. The key insight is that this refinement enhances the efficiency and robustness of the design. Specifically, even when seasonality is weaker than anticipated (e.g., a less pronounced daily pattern compared to the variation in hidden states), such temporal coupling leverages the serial autocorrelation between local periods of the dynamic system, creating an additional layer of balance to the original design.

To study the performance of our new designs, we conduct numerical simulations where we implement and compare other benchmarks in the literature. We have two different simulations, one is a time series model-based setting and the other is a more realistic case based on a ride-sharing simulator from Farias et al. (2022). We show that our new designs achieve a better bias-variance tradeoff, and particularly, a smaller RMSE than existing approaches.

Finally, we emphasize that the features of our design and analysis are shaped by practical considerations, making them not only valuable for switchback experiments but also broadly applicable to the wider landscape of experimental design under interference.

1.3. Related Works

Besides the literature in switchback experiments, this paper is broadly related to experimentation in business under interference – a critical area in experimental design and causal inference, getting particularly more attention in operations context. Improving the performance of such experiments has been approached from two perspectives.

The first perspective is optimizing experimental design with standard estimators. Researchers have focused on designing experiments that account for interference while still using standard estimators like the difference-in-means or inverse probability weighting (IPW). The goal is to structure the experiment in a way that minimizes bias and variance from these estimators. The general principle for addressing population interference is to properly aggregate units into clusters (Ugander et al. 2013, Basse and Feller 2018, Ugander and Yin 2020, Leung 2020, 2022). By randomizing over clusters instead of individual units, dependence between units caused by network connections is reduced without losing much power. Viviano et al. (2023) and Zhao (2024b) further formulate the problem into discrete optimizations and figure out computational strategies. In the case of temporal interference with multiple units, different treatment allocation strategies are studied (Xiong et al. 2019, Basse et al. 2023, Ni et al. 2023, Missault et al. 2025). We refer to Zhao (2024a) for a more comprehensive tutorial on optimizing experimental designs.

The second perspective is developing better estimators with standard design of experiments. Researchers have developed estimators that exploit the structural properties of the interference, often under standard Bernoulli designs where treatments are assigned independently across units. A number of studies (Bright et al. 2022, Dhaouadi et al. 2023, Johari et al. 2022, Li et al. 2021, Si 2023, Li et al. 2023, Zhu et al. 2024, Weng et al. 2024, Zhan et al. 2024, Chen et al. 2025, Zhang et al. 2025) focus on interference that occurs when experimental units share a common competitive resource, often modeled as demand/supply constraints. They typically leverage the one-sided or two-sided structure to reduce the bias caused by interference. When the system is dynamic, the Markovian property is used to better manage the bias-variance tradeoff (Farias et al. 2022, Hu and Wager 2022, Li et al. 2023, Jia et al. 2024, Chen et al. 2024). Besides, Shirani and Bayati (2023) examine the estimation of treatment effect in large random graphs using causal message passing. Munro et al. (2021) and Wager and Xu (2019) suggest using an equilibrium price to capture interference in large markets.

Our work is primarily aligned with the first perspective—enhancing the efficiency and robustness of experimental designs. Although our methods are complemented by non-parametric inference techniques—including a central limit theorem and a variance estimator—the main thrust of the paper lies in providing a powerful design framework that pre-empts challenges typically encountered in switchback experiments with carryover effects. This places our paper squarely within the literature

on optimizing switchback experiments, offering actionable strategies that harmonize theoretical insights with practical implementation.

2. Problem Setup

2.1. Assignment Path and Potential Outcomes

Suppose that there are T time periods. At each time point $t \in [T] = \{1, 2, \ldots, T\}$, we assign the unit to receive an intervention $W_t \in \{0, 1\}$. For example, one experimental period could be one hour and T could be two weeks, i.e., $T = 336$ when one period is one hour. Following convention, we say that the unit is assigned to treatment if $W_t = 1$ and control when $W_t = 0$. The assignment path is then the collection of assignments and is denoted using a vector notation whose dimensions are specified in the subscript, $\boldsymbol{W}_{1:T} = (W_1, W_2, \ldots, W_T) \in \{0, 1\}^T$. We adopt the convention that $\boldsymbol{W}_{1:T}$ stands for a random assignment path, while $\boldsymbol{w}_{1:T}$ stands for one realization. The design of switchback experiments induces a probabilistic distribution over assignment paths $\boldsymbol{w}_{1:T} \in \{0, 1\}^T$. Formally, a design of switchback experiment is any $\eta : \{0, 1\}^T \to [0, 1]$ such that

$$\sum_{\boldsymbol{w}_{1:T} \in \{0,1\}^T} \eta\left(\boldsymbol{w}_{1:T}\right) = 1, \quad \eta\left(\boldsymbol{w}_{1:T}\right) \geq 0, \forall \boldsymbol{w}_{1:T} \in \{0, 1\}^T.$$

Following the extended potential outcomes framework, at time $t \in [T]$, we posit that for each possible assignment path $\boldsymbol{w}_{1:T}$ there exists a corresponding potential outcome denoted by $Y_t\left(\boldsymbol{w}_{1:T}\right)$, which could be different KPIs during one experimental period. The set of all potential outcomes is collected in $\mathbb{Y} = \left\{Y_t\left(\boldsymbol{w}_{1:T}\right)\right\}_{t \in [T], \boldsymbol{w}_{1:T} \in \{0,1\}^T}$ with support $\mathbb{Y} \in \mathcal{Y}$. We adopt the design-based perspective (Neyman 1923, Fisher 1936, Kempthorne 1955, Rubin 1980, Imbens and Rubin 2015, Abadie et al. 2020) and treat the potential outcomes $\mathbb{Y}$ as fixed but unknown quantities, or equivalently, we condition on $\mathbb{Y}$.

2.2. Causal Estimand and Estimator

In this paper, we are interested in the average global treatment effect that measures the difference between permanently adopting the treatment versus the control:

$$\tau(\mathbb{Y}) = \frac{1}{T} \sum_{t=1}^{T} \left[Y_t\left(\mathbf{1}\right) - Y_t\left(\mathbf{0}\right)\right], \tag{1}$$

where $\mathbf{1}$ and $\mathbf{0}$ are vectors of length T of all 1s and 0s, respectively. For every experiment conducted, only one assignment path $\boldsymbol{w}_{1:T}^{\text{obs}}$ is realized following the design η. Accordingly, the potential outcomes $Y_t(\boldsymbol{w}_{1:T}^{\text{obs}}), \forall t \in [T]$ are observed, but all the potential outcomes under other assignment paths cannot be observed (Holland 1986). Consequently, making inference for $\tau(\mathbb{Y})$ is extremely hard (Basse and Airoldi 2018).

Given the challenge of identifying $\tau(\mathbb{Y})$ directly, a common strategy is to develop a causal estimator that approximates $\tau(\mathbb{Y})$. To achieve this, we first introduce the non-anticipating assumption (Bojinov and Shephard 2019).

ASSUMPTION 1. *For any $t \in [T]$ and any two assignment paths $\boldsymbol{w}_{1:T}, \boldsymbol{w}'_{1:T} \in \{0,1\}^T$ with $\boldsymbol{w}_{1:t} = \boldsymbol{w}'_{1:t}$, we have $Y_t(\boldsymbol{w}_{1:T}) = Y_t(\boldsymbol{w}'_{1:T}) = Y_t(\boldsymbol{w}_{1:t})$.*

This assumption simplifies the structure of carryover effects by ensuring that the potential outcome at time t depends only on past treatment assignments. However, it still permits an infinite order of carryover effects—even the first treatment assignment w_1 could impact the potential outcome at the last period T. Under this assumption, a practical estimation strategy is to examine the treatment assignments in the recent past—for instance, the past period along with the current period. Specifically, once two consecutive periods are in the treatment group, we use the observed outcome $Y_t(\boldsymbol{W}_{1:t-1}, \mathbf{1}_{t-1:t})$ as a proxy for $Y_t(\mathbf{1})$, with a similar approach for the control group. This reasoning leads to the following truncated Horvitz-Thompson estimator[1]:

$$\widehat{\tau}(\boldsymbol{W}_{1:T}; \mathbb{Y}, \eta) = \frac{1}{T} \sum_{t=1}^{T} \left\{ Y_t(\boldsymbol{W}_{1:t-2}, \mathbf{1}_{t-1:t}) \frac{\mathbb{1}\{\boldsymbol{W}_{t-1:t} = \mathbf{1}_2\}}{\mathbb{P}(\boldsymbol{W}_{t-1:t} = \mathbf{1}_2)} - Y_t(\boldsymbol{W}_{1:t-1}, \mathbf{0}_{t-1:t}) \frac{\mathbb{1}\{\boldsymbol{W}_{t-1:t} = \mathbf{0}_2\}}{\mathbb{P}(\boldsymbol{W}_{t-1:t} = \mathbf{0}_2)} \right\} \tag{2}$$

Notice that this estimator is unbiased for $\tau(\mathbb{Y})$ when the carryover effects are bounded to one period. Otherwise, the estimator will be biased as the potential outcome depends on assignments before $t-1$; if the dependence is strong, then the gap between $Y_t(\boldsymbol{W}_{1:t-1}, \mathbf{1}_{t-1:t})$ and $Y_t(\mathbf{1})$ could be substantial. To this end, a more general estimator $\widehat{\tau}^{(m)}(\boldsymbol{W}_{1:T}; \mathbb{Y}, \eta)$ could be defined by checking every latest $m \geq 2$ treatment assignments (e.g., $\mathbb{1}\{\boldsymbol{W}_{t-m+1:t} = \mathbf{1}_m\}$). The choice of m naturally leads to a bias-variance trade-off – although larger m generates better proxy with low bias, fewer numbers of proxies would be available given a fixed experimental horizon, causing high variance. [2] Analyzing this tradeoff is intricate, heavily relying on the underlying data generation process. Some similar notions are developed in the literature to study the rate-optimal MSE, such as the duration of burn-in periods in Hu and Wager (2022) and the radius in Jia et al. (2024). Despite their elegant asymptotic theories, in practice, determining the choice of nonasymptotic optimal solution (e.g., optimal burn-in period or optimal radius) in the finite sample regime is still challenging.

In this work, our main focus is on the design of experiments, and we concentrate on the case $m = 2$ as our estimator. In fact, the simplification to $m = 2$ in (2) is largely without loss of generality, because it can be viewed as an equivalent representation by redefining the length of a period. For example, $m = 4$ with one hour per period is equivalent to $m = 2$ with two hours per period, so the critical factor is the duration of the latest four hours, which is specified as a hyper-parameter by the decision maker. We also investigate the impact of defining the period length on empirical performance in the numerical study (Section 7).

[1] We note that the notation $\mathbf{1}_{t-1:t}$ is not rigorous because on the boundary like $t = 1$, the assignments $\boldsymbol{W}_{0:1}$ may not be well defined. In these cases, we introduce a warm-up period before $t = 1$ so the observation starts from $t = 0$ and our estimators are always defined on periods 1 to T.

[2] This general estimator is actually unbiased for a different casual estimand defined in Bojinov and Shephard (2019), which quantifies the effect of being treated for m periods.

3. Model-assisted Design of Switchback Experiments

In this section, we develop a new family of model-assisted switchback designs, particularly driven by time series models.

3.1. Blocking with Time Series Model: Seasonality and Auto-correlation

The key feature of switchback experiments is that experimental units are defined by time periods, and samples are collected sequentially over time. Given historical data preceding the experiment, we can fit a general time-series model to uncover dynamic structures that inform experimental design. For illustration, consider the following Bayesian structural time-series model (Scott and Varian 2014, Brodersen et al. 2015):

$$\begin{aligned} y_t &= \alpha_t + s_t + \varepsilon_t, \\ \alpha_t &= \gamma\alpha_{t-1} + \beta_t + \eta_t, \\ \beta_t &= \theta\beta_{t-1} + \xi_t, \end{aligned} \tag{3}$$

where α_t represents a latent state that evolves dynamically, and s_t captures cyclical seasonal effects. The latent state α_t specifies an unobserved level component that follows an autoregressive process with the parameter γ. It also includes a local trend component β_t that follows an autoregressive process itself with the parameter θ. We have three independent residual noises ε_t, η_t and ξ_t. [3]

The calibration of such models often reveals the seasonal pattern s_t of the time series such as daily cycles, whose identification is often robust. Even if model (3) is misspecified (e.g., in terms of residual distributions or the autoregressive structure), consistent estimation of seasonal patterns is often still achievable (Brockwell 1991). Also, the intervention itself is unlikely to meaningfully distort these long-term seasonal structures. While it may affect the strength of correlations across periods, these effects are typically second-order and do not erase the underlying patterns. For example, to allow for the intervention, we can extend the model as follows:

$$\begin{aligned} Y_t(\boldsymbol{w}_{1:t}) &= \alpha_t + s_t + \lambda_t^Y w_t + \varepsilon_t, \\ \alpha_t &= \gamma\alpha_{t-1} + \lambda_t^\alpha w_t + \beta_t + \eta_t, \\ \beta_t &= \theta\beta_{t-1} + \xi_t, \end{aligned} \tag{4}$$

where $\lambda_t^Y w_t$ represents the immediate impact of treatment on the observed outcome, and $\lambda_t^\alpha w_t$ captures its influence on the latent state – causing the carryover effect.

Our first key design principle is to balance treatment and control assignments across the seasonal cycles identified from historical data—what we call *blocks*. For example, if each time period represents one hour, then fitting a time series[4] using the NYC taxi data reveals a daily periodicity, so we define

[3] Observed covariates (e.g., weather conditions) can be incorporated via a term in y_t such as $\boldsymbol{\beta}_t'\boldsymbol{x}_t$, though we omit them here for simplicity since they are not the focus of our design.

[4] We will see more details in the numerical study in Section 7.

$B = 24$ blocks. We aim to ensure that, for each specific block (e.g., 7–8 pm), the number of treatment and control observations is equal across the $K = T/24$ days, where T is the total number of periods. This can be achieved via complete randomization within each block of size K, independently across the B blocks—i.e., a randomized block design (Wilk 1955).

However, two concerns arise with this standard block-based approach: (i) In the estimation procedure, not all periods can be used. For instance, when the assignment switches from period $t-1$ to t, such as $\boldsymbol{w}^{\text{obs}}_{t-1:t} = (1,0)$, the data at t will be discarded to avoid contamination. In extreme cases, if every day's realized assignment at 6–7 pm (w^{obs}_t) is followed by the opposite assignment at 7–8 pm ($w^{\text{obs}}_{t+1} = 1 - w^{\text{obs}}_t$), no observation would be available for 8 pm, harming the reliability of the experiment. (ii) Besides the seasonality, another prominent feature of general time series models is serial autocorrelation that characterizes the temporal dependency of observations, for example, reflected by the auto-regressive parameters γ and θ in the time series (4). While balancing within blocks accounts for seasonality, it may not be robust against other exogenous factors (e.g., weather, competitor actions, events) impacting the time series. That is, residual variability (e.g., from η_t and ξ_t in (4)) could be substantial, amplifying the serial autocorrelation and reducing the value gained from within-block balancing.

3.2. The New Design with Adaptive Block Randomization

We propose a new design η to overcome the concerns above, and later we will extend it in Section 5 for better addressing the second concern. The detailed steps of the design are presented in Algorithm 1. Here, we elaborate on the key ideas behind it.

We first introduce a parameter $K_v \in 2\mathbb{Z} \le K$ and $\rho = K_v/K \in [0,1]$. We let $\mathcal{T}_b = \{b, b+B, b+2B, \ldots\}$ denote the indices of time periods in the block $b \in [B]$ and $\boldsymbol{W}_b^B = \{W_t, \forall t \in \mathcal{T}_b\}$ denote the assignments in the block $b \in [B]$. The new design generates the assignments progressively from the first block $b = 1$ to the last block $b = B$ (i.e., from $\boldsymbol{W}_1^B$ to $\boldsymbol{W}_B^B$) in an adaptive manner. Conditioning on the assignments of a block b, we randomly generate the assignments of the next block $b+1$ in a way that guarantees we observe $K_v/2$ number of $(1,1)$ and $(0,0)$, and $(K-K_v)/2$ number of $(1,0)$ and $(0,1)$ in these two consecutive blocks $\boldsymbol{W}^B_{b:b+1} = \{\boldsymbol{W}_{t:t+1}, \forall t \in \mathcal{T}_b\}$. Precisely, for $t+1 \in \mathcal{T}_{b+1}$, we completely randomize the assignments following

$$\begin{aligned} \mathbb{P}(W_{t+1} = 1 | W_t = 1) = \mathbb{P}(W_{t+1} = 0 | W_t = 0) = \rho, \\ \mathbb{P}(W_{t+1} = 0 | W_t = 1) = \mathbb{P}(W_{t+1} = 1 | W_t = 0) = 1 - \rho. \end{aligned} \tag{5}$$

Consequently, for any feasible assignment path, we collect $\rho K/2$ valid observations for both treatment and control in each block (e.g., every hour of a day). The performance of the design is determined by the parameter ρ — a small ρ collects insufficient data while a large ρ collects highly dependent data across blocks. The following theorem formalizes the key property of our new design, which plays an important role in the variance and subsequent inference.

Algorithm 1 The New Design η

1: Input: K, $2 \le K_v \le K (\rho = K_v/K)$.
2: Generate the assignments of the first block $\boldsymbol{W}_1^B$ following a complete randomization with $K/2$ treatments and $K/2$ controls.
3: **for** $b \in \{2, \ldots, B\}$ **do**
4: Let $\mathcal{T}_{b-1}^1$ be the set of periods where for $t \in \mathcal{T}_{b-1}^1$, there exists some k such that $t = (b-1) + (k-1)B$ and $W_t = 1$; Let $\mathcal{T}_{b-1}^0$ be the set of periods where for $t \in \mathcal{T}_{b-1}^0$, there exists some k such that $t = (b-1) + (k-1)B$ and $W_t = 0$.
5: To generate the assignments $W_{t+1}, \forall t \in \mathcal{T}_{b-1}^1$, we create a set that contains $K_v/2$ treatments and $(K - K_v)/2$ controls
6: Conduct a random permutation with $\mathbb{P}(W_{t+1} = 1 | W_t = 1) = \rho$ and $\mathbb{P}(W_{t+1} = 0 | W_t = 1) = 1 - \rho$.
7: We symmetrize Step 5 - 6 for $W_{t+1}, \forall t \in \mathcal{T}_{b-1}^0$.
8: **end for**
9: Let $W_0 = W_T$ and return the assignment path $\boldsymbol{W}_{0:T}$.

THEOREM 1. *Let S be the set of following K assignment pairs:*

$$S = \left\{ \underbrace{(1,1), \ldots, (1,1)}_{K_v/2}, \underbrace{(0,0), \ldots, (0,0)}_{K_v/2}, \underbrace{(1,0), \ldots, (1,0)}_{(K-K_v)/2}, \underbrace{(0,1), \ldots, (0,1)}_{(K-K_v)/2} \right\}.$$

Under the design η, for block $b > 1$, the marginal distribution of the treatment assignments $\boldsymbol{W}_{b-1,b}^B$ follows a random permutation over S. That is, for $t \in \mathcal{T}_b$,

$$\mathbb{P}(W_{t-1:t} = (1,1)) = \mathbb{P}(W_{t-1:t} = (1,1)) = 0.5\rho,$$
$$\mathbb{P}(W_{t-1:t} = (1,0)) = \mathbb{P}(W_{t-1:t} = (0,1)) = 0.5(1-\rho).$$

We leave the proof in Appendix A.

To better illustrate the randomization mechanism of the new design, we visualize in Table 3 the process of one realization with $K = 8$ and $\rho = 0.5$. We first look at the block $b = 7$. By checking the assignments $\boldsymbol{W}_{6:7}^B$, we see that it is exactly a permutation of set S. Now we move forward to $b = 8$ based on the realized assignments $\boldsymbol{W}_7^B$. All 8 rows are split into treatment (shaded) and control groups. For the treatment group (i.e. $k = 1, 2, 4, 6$), we randomly shuffle $\rho K/2 = 2$ treatments and $(1-\rho)K/2 = 2$ controls to generate the corresponding assignments at the block $b = 8$; for the control group (i.e. $k = 3, 5, 7, 8$), we randomly shuffle $\rho K/2 = 2$ controls and $(1-\rho)K/2 = 2$ treatments. After that, by checking the assignments $\boldsymbol{W}_{7:8}^B$, we see that it still follows a permutation of set S.

Before proceeding to the analysis of our new design. we discuss the design parameter ρ and its relationship with prior works. Early works with a similar estimator studied the optimal switchback

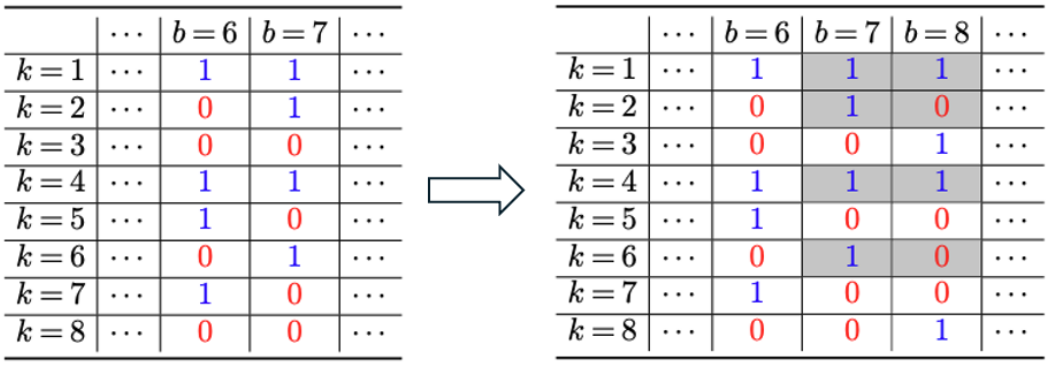


Figure 3 **Illustration of adaptive block randomization**

design within certain families, such as the Bernoulli design Bojinov et al. (2022), Hu and Wager (2022), Jia et al. (2024) and the complete randomized design Ni et al. (2023). One main drawback of their practical adoptions is the random effective sample size — making the estimation less stable. Our new design resolves this issue by specifying a fixed number of ρK effective samples. We particularly highlight two special cases $\rho = 1$ and $\rho = 0.5$. When $\rho = 1$, the design requires minimal switching, except for daily adjustments, so the propensity score is nearly 0.5, close to the setting without carryover effects with a maximal number of observations. This configuration maximizes within-block balance, fully leveraging seasonality; however, it also becomes more sensitive to temporal dependence across blocks, as fewer switches limit the design's ability to decorrelate treatment from evolving system dynamics. When $\rho = 0.5$, the propensity score $\mathbb{P}(\boldsymbol{W}_{t-1:t} = (1,1)) = 0.25$ is the same as the one from the Bernoulli design in Bojinov et al. (2022), meaning that, on average, both designs yield the same number of effective samples. However, our design differs by cutting off the tails through the use of seasonal structure in the time series. Specifically, if fewer effective samples are collected during the early days, the design self-compensates by allocating more effective samples in later days, maintaining balance over the entire cycle. While this configuration discards half of the observations—sacrificing some within-block sample efficiency—it provides stronger protection against temporal dependence through more frequent switches across blocks. In this paper, we are particularly interested in analyzing the performance of the design by choosing $\rho \in [0.5, 1]$. Note that $\rho = 0.5$ essentially flips a fair coin between treatment and control for every new assignment, regardless of the past assignment. Letting $\rho < 0.5$ enforces the randomization to be more likely to produce consecutive, alternating assignments (e.g., $\boldsymbol{w}_{t-5:t} = (0,1,0,1,0,1)$), which is results in an excessive discarding of data.

Remark. Our design is model-assisted, which means that we leverage the structure of the time series, without relying on the estimated parameters of the model. It ensures that even if the time series model is misspecified, as long as the periodicity is correctly identified, our design remains

beneficial. In the worst-case scenario where even the periodicity is incorrect, the design still functions as a valid design, maintaining balance across treatment and control groups over the experimental horizon.

4. Analysis of Bias and Variance

In this section, we investigate the properties of the model-assisted design by analyzing its bias and variance.

4.1. Bias Analysis

To start with, let us revisit the estimator

$$\widehat{\tau}\left(\boldsymbol{W}_{1:T};\mathbb{Y},\eta\right)=\frac{1}{T}\sum_{t=1}^{T}\left\{Y_t\left(\boldsymbol{W}_{1:t-2},\mathbf{1}_{t-1:t}\right)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=\mathbf{1}_2\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=\mathbf{1}_2\right)}-Y_t\left(\boldsymbol{W}_{1:t-1},\mathbf{0}_{t-1:t}\right)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=\mathbf{0}_2\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=\mathbf{0}_2\right)}\right\}. \tag{6}$$

Its expectation is denoted by $\bar{\tau}(\mathbb{Y},\eta)=\mathbb{E}[\widehat{\tau}\left(\boldsymbol{W}_{1:T};\mathbb{Y},\eta\right)]$. The bias of the estimator can be characterized as

$$\begin{aligned}\tau(\mathbb{Y})-\bar{\tau}(\mathbb{Y},\eta)=&\frac{1}{T}\sum_{t=1}^{T}\left\{Y_t(\mathbf{1})-\sum_{\boldsymbol{w}_{1:t-2}}Y_t\left(\boldsymbol{w}_{1:t-2},1,1\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t}=(\boldsymbol{w}_{1:t-2},1,1)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}\right\}\\&+\frac{1}{T}\sum_{t=1}^{T}\left\{\sum_{\boldsymbol{w}_{1:t-2}}Y_t\left(\boldsymbol{w}_{1:t-2},0,0\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t}=(\boldsymbol{w}_{1:t-2},0,0)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(0,0)\right)}-Y_t(\mathbf{0})\right\}.\end{aligned} \tag{7}$$

Without further assumptions, the bias from ignoring the carryover effects can be arbitrarily large. In most practical applications, however, the influence of past treatments should diminish as the time between the assignment and the outcome grows. The assumption below captures this intuition and enables us to quantify and bound the bias.

ASSUMPTION 2. *For $t\in[T]$, let $\phi_t(q)$ be a decreasing function in $q\geq 1$, such that*

$$\max_{\boldsymbol{w}_{1:t-q},\boldsymbol{w}'_{1:t-q}}|Y_t(\boldsymbol{w}_{1:t-q},\boldsymbol{d})-Y_t(\boldsymbol{w}'_{1:t-q},\boldsymbol{d})|\leq\phi_t(q),\forall\boldsymbol{d}\in\{0,1\}^q. \tag{8}$$

This assumption is similar to the spatial interference requirement in Leung (2022). The function $\phi_t(q)$ measures the degree of carryover effects in terms of the maximum change to period t's potential outcome caused by manipulating treatments assigned to ones that are at least q periods away. The decreasing feature of $\phi_t(q)$ requires the impact of carryover effects to vanish with q so that treatments assigned to more distant periods interfere less.

In this paper, we do not assume a certain order of decay for $\phi_t(p)$. As mentioned early, our goal is not to derive results for rate-optimal MSE, but to investigate how bias and variance are shaped by the design parameter. To this end, we first present the following theorem:

THEOREM 2. *Under Assumption 2, the bias has an upper bound decreasing in ρ:*

$$|\tau(\mathbb{Y}) - \bar{\tau}(\mathbb{Y}, \eta)| \leq \frac{2}{T} \sum_{t=1}^{T} \left(\rho \phi_t(3) + (1-\rho)\phi_t(2)\right). \tag{9}$$

We leave the proof in Appendix B. The intuition behind the monotonicity of ρ on bias is that if the design enforces more effective samples (i.e., $\boldsymbol{w}_{t-1:t}^{\text{obs}} \in \{(1,1),(0,0)\}$) with a large ρ, the randomization mechanism has to switch less frequently between treatment and control. This leads to more occurrences of long consecutive treatment/control assignments, whose potential outcomes are closer to the one under global treatment/control in $\tau(\mathbb{Y})$. We remark that a similar property also holds in regular Bernoulli design of switchback experiments (Bojinov et al. 2022, Hu and Wager 2022, Jia et al. 2024), where larger window size for switching increases the propensity score and reduces the bias.

To better understand the magnitude of potential bias in estimation, we revisit the time-series model in (4) to characterize its implications. If the data are indeed generated according to this model, the delayed impact of treatment on the latent state α_t propagates over time through the autoregressive structure. In this way, we have

$$\phi_t(q) = \sum_{p \geq q} |\lambda_{t-p}^{\alpha}| \gamma^p. \tag{10}$$

The bias is then

$$|\tau(\mathbb{Y}) - \bar{\tau}(\mathbb{Y}, \eta)| \leq \frac{2}{T} \sum_{t=1}^{T} \left(\sum_{p \geq 3} |\lambda_{t-p}^{\alpha}| \gamma^p + (1-\rho)|\lambda_{t-2}^{\alpha}| \gamma^2 \right). \tag{11}$$

It implies an exponential decay of carryover effects if $|\gamma| < 1$. This is known as the rapid mixing condition[5] inferred from the autoregressive part of the model. This condition is common in two-sided platforms with complex dynamic systems where a hidden state reflects some supply-demand status, with the intervention being pricing/matching adjustment in ridesharing, new order assignment in food delivery, etc. This bias bound offers a useful pre-experiment check. Even though we cannot identify the time-wise treatment effect on the latent state λ_t^{α} at each period t, we can still assess how much of that effect the estimator may omit. Specifically, its contribution to the bias is bounded by $2|\lambda_t^{\alpha}|((1-\rho)\gamma^2 + \sum_{p \geq 3} \gamma^p)$, where the total indirect effect is $|\lambda_t^{\alpha}| \sum_{p \geq 0} \gamma^p$. Taking the ratio between them yields:

$$\frac{2((1-\rho)\gamma^2 + \sum_{p \geq 3} \gamma^p)}{\sum_{p \geq 0} \gamma^p} \approx 2(\rho\gamma^3 + (1-\rho)\gamma^2). \tag{12}$$

[5] Such a condition is also discussed in Hu and Wager (2022) and Jia et al. (2024) under Markov systems. One key difference is that the mixing condition defined there is from a sampling-based perspective while this paper studies the experiments from a design-based perspective.

This expression quantifies the fraction of the carryover effect omitted by the estimator. In practice, by fitting the time series model to pre-experimental data, we can estimate $\widehat{\gamma}$ and plug it into the formula to assess the impact of the potential bias. If $\widehat{\gamma}$ is large, it may be advisable to increase the duration of each time period (e.g., switching from hourly to two-hour interval), refit the time series model until the impact of bias becomes tolerable.

4.2. Variance Analysis

4.2.1. Auxiliary Notations To derive the variance of our design, note that the observations from different blocks are naturally correlated given how assignments are block-wise generated and the existence of carryover effects. It is also intuitive that such correlation may decrease as the distance between two blocks increases, which needs to be characterized precisely for understanding the variance. To this end, we need to introduce some additional notations. First, we define the overlapping indices by

$$\begin{aligned}\gamma^{+}_{t,t'} &= \frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1)|\boldsymbol{W}_{t-1:t}=(1,1))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1))} = \frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0)|\boldsymbol{W}_{t-1:t}=(0,0))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0))},\\ \gamma^{-}_{t,t'} &= \frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0)|\boldsymbol{W}_{t-1:t}=(1,1))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0))} = \frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1)|\boldsymbol{W}_{t-1:t}=(0,0))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1))}.\end{aligned} \tag{13}$$

The index $\gamma^{+}_{t,t'}$ measures how much knowing that the period t is in a valid treatment (control) condition changes the likelihood of observing that same pattern in the period t', relative to its marginal probability. In contrast, $\gamma^{-}_{t,t'}$ measures this dependency for the opposite pattern. In essence, these indices capture the strength of the association between the assignments in two different periods. Following the definition, we note that under our experimental design, for any two blocks b and b' with distance $\delta_{b,b'}=|b'-b|$, $\gamma^{+}_{t,t+\delta_{b,b'}}$ is invariant for all time periods $t\in\mathcal{T}_b$. With a slight abuse of notation, we henceforth denote this common value by $\gamma^{+}_{b,b'}\equiv\gamma^{+}_{t,t+\delta_{b,b'}}$, and it will be used in later analysis to represent the association between the assignments in two blocks b and b'.

Next, we characterize a similar association between the outcomes. We let $\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]$ be the expected treatment outcome at period $t\in\mathcal{T}_b$ over random assignments $\boldsymbol{W}_{1:t-2}$ conditioning on $\boldsymbol{W}_{t-1:t}=(1,1)$ and let $\overline{\mathbb{E}[Y_b(\cdot,1,1)]}=\frac{1}{K}\sum_{t\in\mathcal{T}_b}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]$ be the average expected treatment outcome in the block b. For another block b', we let $\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]$ be the conditional expected treatment outcome at period $t+\delta_{b,b'}\in\mathcal{T}_{b'}$. Now we introduce the following notions: for $u\in\{0,1\},v\in\{0,1\}$,

$$\begin{aligned}S^{u,v}_{b,b'} = \frac{1}{K-1}\sum_{t\in\mathcal{T}_b}&(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},\boldsymbol{u}_2)]-\overline{\mathbb{E}[Y_b(\cdot,\boldsymbol{u}_2)]})\\ &\cdot(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},\boldsymbol{v}_2|\boldsymbol{W}_{t-1:t}=\boldsymbol{u}_2)]-\overline{\mathbb{E}[Y_{b'}(\cdot,\boldsymbol{v}_2|\boldsymbol{W}_{t-1:t}=\boldsymbol{u}_2)]}).\end{aligned} \tag{14}$$

Essentially, $S^{u,v}_{b,b'}$ characterizes the sample covariance of the outcomes given $\boldsymbol{W}_{t-1:t}=\boldsymbol{u}_2,\boldsymbol{W}_{t+\delta_{b,b'}-1:t+\delta_{b,b'}}=\boldsymbol{v}_2$ between two blocks b and b'. It reduces to the sample variance of expected

outcomes within a certain block when $b = b'$ (with a simplified notation S_b^u). Moreover, we introduce for $u \in \{0,1\}, v \in \{0,1\}$,

$$\sigma_{t,t'}^{u,v} = \mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2}, \boldsymbol{u}_2), Y_{t'}(\boldsymbol{W}_{1:t'-2}, \boldsymbol{v}_2)\right). \tag{15}$$

Basically, $\sigma_{t,t'}^{u,v}$ characterizes the sample conditional covariance of outcomes at any two periods t and t' given $\boldsymbol{W}_{t-1:t} = \boldsymbol{u}_2, \boldsymbol{W}_{t'-1:t'} = \boldsymbol{v}_2$. It reduces to the sample conditional variance when $t = t'$ (with a simplified notation σ_t^u).

4.2.2. Variance Decomposition Now we are ready to decompose the variance of the estimator into two parts by conditioning on $\boldsymbol{W}_{t-1:t}$ for each period t. The following theorem gives the precise characterization for the variance.

THEOREM 3. *The variance of the estimator* (2) *under the design* η *satisfies*

$$\begin{aligned}
&\mathrm{Var}_\eta(\widehat{\tau}) = V_1 + V_2 \\
&V_1 = \frac{1}{B^2 K} \sum_{b,b' \in [B]} (\gamma_{b,b'}^+ (S_{b,b'}^{1,1} + S_{b,b'}^{0,0}) - \gamma_{b,b'}^- (S_{b,b'}^{1,0} + S_{b,b'}^{0,1})) - \frac{1}{K(K-1)} \sum_{k=1}^{K} (\bar{\tau}_k - \bar{\tau})^2 \\
&V_2 = \frac{1}{B^2 K^2} \sum_{b,b' \in [B]} \sum_{t \in \mathcal{T}_b, t' \in \mathcal{T}_{b'}} (\gamma_{t,t'}^+ (\sigma_{t,t'}^{1,1} + \sigma_{t,t'}^{0,0}) - \gamma_{t,t'}^- (\sigma_{t,t'}^{1,0} + \sigma_{t,t'}^{0,1}))
\end{aligned} \tag{16}$$

where $\bar{\tau}_k$ *refers to the average treatment effect in the* k*-th day and* $\bar{\tau}$ *is the expectation of* $\widehat{\tau}$*.*

The proof is given in Appendix C. The first part V_1 quantifies the variation in the expected outcomes conditional on the recent history $\boldsymbol{W}_{t-1:t}$. It is driven by the sample covariances $S_{b,b'}^{u,v}$ computed between every pair of blocks, weighted by the overlapping indices $\gamma_{b,b'}^+$ and $\gamma_{b,b'}^-$ between these blocks. In essence, V_1 generalizes the variance expression found in completely randomized designs (see, e.g., Neyman (1923) and Li (2017)). The first part V_2 captures the variation attributable to the earlier assignments $\boldsymbol{W}_{1:t-2}$. Here, the covariances $\sigma_{t,t'}^{u,v}$ play a similar role as in V_1, which are also weighted by $\gamma_{t,t'}^+$ and $\gamma_{t,t'}^-$. The value of overlapping indices themselves can be derived recursively based on our design process (see (67) - (69)).

Despite the added complexity in the variance compared to classical randomized experiments, we demonstrate in Section 6 that a variance estimator can be developed for inference. Nevertheless, we are interested in understanding how the experimental design—in particular, the design parameter ρ—affects the variance. We have observed that a larger value of ρ leads to a smaller bias, suggesting a natural conjecture that it results in a larger variance, in line with the typical bias-variance trade-off. However, this relationship is not straightforward in (16) since every component of the variance is a function of ρ. To gain clearer analytical insights, we must make some simplifications and approximations.

4.2.3. Variance Approximation and Reformulation Our strategy is to isolate the part of the variance that depends solely on ρ and approximate the remainder. For the overlapping indices $\gamma^+_{t,t'}$ and $\gamma^-_{t,t'}$, we can control their dependence on ρ as we govern the design process. In contrast, the elements in $\boldsymbol{S}$ and $\boldsymbol{\sigma}$ are more implicit, being jointly affected by the design parameter ρ and potential outcomes. Combining these allows us to obtain an upper bound for V_2.

COROLLARY 1. *Under Assumption 2, with* $\phi(q) = \max_t \phi_t(q)$,

$$\begin{aligned} V_2 \le V_2^{\mathrm{U}} =& \frac{4(1-\rho)}{B(K-1)}\phi(2)^2 + \frac{4(1-\rho^2)}{B(K-2)}\sum_{q=3}^{\infty}\phi(q)^2 \\ &+ \frac{8\rho(1-\rho)}{B^2K}\sum_{b\neq b'\in[B]}\phi(2)\phi(2+\delta_{b,b'}) + \frac{4\rho(1-\rho)(3K/2-2)}{B^2K(K/2-1)}\sum_{b\neq b'\in[B]}\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'}). \end{aligned} \tag{17}$$

The proof is given in Appendix D. The variance upper bound V_2^{U} is decreasing in ρ, mirroring the monotonic behavior of the bias. Intuitively, as ρ approaches 1, there are fewer switches, resulting in limited variation in the assignments $\boldsymbol{W}_{1:t-2}$ that are not being captured in the truncated estimator. One might expect the first component of the variance, V_1, to be increasing with ρ, reflecting the typical bias-variance trade-off; however, this may not hold. To investigate V_1 in greater detail, we introduce the following assumption.

ASSUMPTION 3. *Consider the following definition for the covariance of the treatment effect:*

$$\begin{aligned} S^{\tau}_{b,b'} = \tfrac{1}{(K-1)}\sum_{t\in\mathcal{T}_b}&(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)] - \mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)] - (\overline{\mathbb{E}[Y_b(\cdot,1,1)]} - \overline{\mathbb{E}[Y_b(\cdot,0,0)]})) \\ &\cdot(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},1,1)] - \mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},0,0)] - (\overline{\mathbb{E}[Y_{b'}(\cdot,1,1)]} - \overline{\mathbb{E}[Y_{b'}(\cdot,0,0)]})). \end{aligned}$$

We assume that the equality $S^{1,0}_{b,b'} + S^{0,1}_{b,b'} = S^{1,1}_{b,b'} + S^{0,0}_{b,b'} - S^{\tau}_{b,b'}$ *holds.*

This assumption re-expresses the covariance of the cross-terms (i.e., the interaction between treatment and control outcomes $S^{1,0}_{b,b'} + S^{0,1}_{b,b'}$) into the forms of the covariances of the treatment outcomes $S^{1,1}_{b,b'}$, the control outcomes $S^{0,0}_{b,b'}$, and the treatment effect itself $S^{\tau}_{b,b'}$. This equality is a standard device in the analysis of completely randomized designs that do not exhibit carry-over effects (Neyman 1923, Imbens and Rubin 2015). The chief hurdle in defending the assumption is that, for any two periods t and $t+\delta_{b,b'}$ drawn from different blocks, the outcome at $t+\delta_{b,b'}$ can in principle depend on the treatment assigned at t through an arbitrary function. Appendix E works through the formal details; here we highlight the conditions under which the assumption is tenable. First, the equality is exact when carryover lasts at most one period, so that once we condition on the two-period assignment vector $\mathbf{W}_{t-1:t}$, no hidden dependence remains; the derivation then reduces to the familiar no-interference case (Bojinov et al. 2022, Chen and Simchi-Levi 2023, Ni et al. 2023). Moreover, a weaker requirement is that the treatment effect in every period t does not depend on assignments

prior to $t-1$. Earlier assignments may still shift both treated and control potential outcomes by the same constant—say, $W_{t-4}=1$ uniformly elevates $Y_t(\mathbf{W}_{t-2},1,1)$ and $Y_t(\mathbf{W}_{t-2},0,0)$—a property satisfied by many generalized linear outcome models but violated by more intricate dynamics, such as habituation effects that involve interactions among past treatments. In fact, period-wise independence is not necessary; it suffices that all periods within a block respond in the same way, so that, for example, the influence of treating 4 p.m. on the 8 p.m. effect is invariant across days. Despite the breadth of these conditions, we acknowledge that they may not hold perfectly in real-world settings. Nonetheless, the resulting discrepancy can be bounded by exploiting $\phi_t(q)$ together with the support of the potential-outcome distribution, leading to an upper variance bound analogous to V_2^{U}. For analytical convenience we adopt the idealized assumption in the sequel, which permits a closed-form expression for the first component of the variance.

COROLLARY 2. *Under Assumption 3,*

$$V_1 = \frac{1}{B^2K}\left\{\sum_{b=1}^{B}\left(\frac{2}{\rho}(S_b^1+S_b^0)-S_b^\tau\right)+\sum_{b\neq b'\in[B]}(2\rho-1)^{\delta_{b,b'}-1}\left(2(S_{b,b'}^{1,1}+S_{b,b'}^{0,0})-S_{b,b'}^\tau\right)\right\}. \tag{18}$$

The proof is given in Appendix E. The structure of this variance decomposition closely resembles that in classical complete randomization (Neyman 1923, Imbens and Rubin 2015, Li and Ding 2017), from which the variance from a random permutation of K assignments within a block is given by $2(S_b^1+S_b^0)-S_b^\tau$. In our framework, however, the design parameter ρ restricts the number of valid observations, thereby adjusting the variance to $\frac{2}{\rho}(S_b^1+S_b^0)-S_b^\tau$. Moreover, this extends to the block-wise covariance representation, where each pair is weighted by $(2\rho-1)^{\delta_{b,b'}-1}$.

4.2.4. Design Parameter The reformulation of the variance in (18) allows us to study the impact of the design parameter ρ in a more succinct way. In this expression, the weights ρ^{-1} and $(2\rho-1)^{\delta_{b,b'}-1}$ exert opposing influences on the variance. When ρ is small, the within-block variances dominate due to the relatively few valid observations, which in turn reduces the sampling efficiency for estimating each block. Conversely, when ρ is large, the between-block covariances (i.e., for $\delta_{b,b'}\geq 2$) become more pronounced because different blocks tend to share the same assignments across days. Consequently, ρ governs the magnitude of V_1 by reflecting the degree of similarity among the potential outcomes across blocks — a larger ρ is preferable when outcomes measured at the same hour across different days are highly correlated, and a smaller ρ is suitable when they are less so.

Let us consider two special cases. A design with $\rho=1$ retains the same assignment throughout the entire day, switching only from day to day. This approach is aggressive, and the variance can become extremely large in certain scenarios. For example, if the outcome at 7 pm can be highly correlated with outcomes from 3 to 11 pm on the same day, grouping all these periods together may

induce substantial variation. Conversely, a design with $\rho = 0.5$ eliminates the positive correlation between blocks that are at least two periods apart (i.e., $2\rho - 1 = 0$). This can be considered as a robust configuration, against arbitrary correlations between distant blocks in the potential outcomes.

Unfortunately, finding the exact optimal choice ρ^* that minimizes $V_1 + V_2^{\mathrm{U}}$ (or even MSE) is extremely challenging. Achieving this would require more assumptions about the outcome models, particularly regarding all covariance terms and $\phi(\cdot)$. Instead, practitioners may choose $\rho \in [0.5, 1]$ based on the domain knowledge of the problem context and pre-experimental data. Here, we provide one empirical strategy for picking an appropriate ρ. Before the experiment, the decision maker often has some historical data for approximating the statistics of the control group $S^{0,0}_{b,b'}$, but has limited knowledge about $S^{1,1}_{b,b'}$ and $S^{\tau}_{b,b'}$. A typical trick is to assume constant treatment effect so that $S^{\tau}_{b,b'} = 0$ and $S^{1,1}_{b,b'} = S^{0,0}_{b,b'}$. Let $\widehat{S}_{b,b'}$ be the estimation of the covariance using historical data. Then we have

$$\widehat{V}_1 = \frac{4}{B^2 K}\left\{\sum_{b=1}^{B}\frac{\widehat{S}_b}{\rho} + \sum_{b\neq b'\in[B]}(2\rho-1)^{\delta_{b,b'}-1}\widehat{S}_{b,b'}\right\}, \tag{19}$$

which is convex in ρ, so it is equivalent to check the first-order condition

$$-\sum_{b=1}^{B}\frac{\widehat{S}_b}{\rho^2} + \sum_{b\neq b'\in[B],\delta_{b,b'}\geq 2} 2(\delta_{b,b'}-1)(2\rho-1)^{\delta_{b,b'}-2}\widehat{S}_{b,b'} = 0, \tag{20}$$

by a line search of $\rho \in [0.5, 1]$. Suppose that $\hat{\rho}$ is the minimizer of $\widehat{V}_1$, then one should choose $\rho \in [\hat{\rho}, 1]$ because the second part of the variance (17) and the bias (9) are decreasing in ρ.

Remark. To guarantee the number of valid observations to be $K_v = \rho K$ at each block, the choice of ρ is actually restricted. Taking $K = 14$ (i.e., two weeks) as instance, ρ can only be chosen from $\{4/7, 5/7, 6/7, 1\}$. Technically, we can extend our design idea to some mixed strategy with a general choice of ρ. That is, we only guarantee $\mathbb{E}[K_v] = \rho K$. For example, $\rho = 4.5/7$ can be characterized by generating $K_v = 8$ valid observations with 50% of chance, and $K_v = 10$ valid observations with 50% of chance. That being said, we do not investigate this extension in detail since it does not add much practical value — the optimal choice ρ^* is unknown and knowing some approximate range is sufficient for most applications, which gives enough flexibility in selecting feasible ρ and K. For example, when $\hat{\rho} \in (4/7, 5/7)$, we could pick $\rho = 5/7$ if the bias is less concerned and otherwise choose $\rho = 6/7$.

5. A Refined Design with Temporal Coupling

The design we introduced earlier offers a new perspective on switchback experiments. We have kept it simple to facilitate clean analysis using only one parameter ρ, though the concerns in Section 3.1 are partially addressed. In this section, we further refine the model-assisted design to achieve better efficiency and robustness.

5.1. Adaptive Block Randomization with Temporal Coupling

Our new design only guarantees $K_v = \rho K$ valid observations over K days, and still admits the assignment path where all valid observations are from certain days and no information is collected from other days. We propose a variant of the new design $\eta^\dagger$ with temporal coupling in Algorithm 2 by adjusting the adaptive block randomization of assignments (i.e. Step 6 in Algorithm 1). Intuitively, when we observe $\boldsymbol{W}_{t-1:t} = (1,1)$, we reduce the chance of assigning treatment to W_{t+1} and when we observe $\boldsymbol{W}_{t-1:t} = (0,1)$, we improve the probability of assigning treatment to W_{t+1}. This eliminates some bad instances with too many switches (e.g., $\boldsymbol{w}_{t-5:t} = (0,1,0,1,0,1)$) and too few switches (e.g., $\boldsymbol{w}_{t-5:t} = (1,1,1,1,1,1)$). As a result, we not only ensure a fixed number of valid observations in each block, but also distribute these observations more evenly across different days. Particularly, we consider the following conditional probabilities for generating the assignments:

$$\begin{aligned}
&\mathbb{P}(W_{t+1}=1|\boldsymbol{W}_{t-1:t}=(1,1)) = \mathbb{P}(W_{t+1}=0|\boldsymbol{W}_{t-1:t}=(0,0)) = \rho^- \le \rho,\\
&\mathbb{P}(W_{t+1}=0|\boldsymbol{W}_{t-1:t}=(1,1)) = \mathbb{P}(W_{t+1}=1|\boldsymbol{W}_{t-1:t}=(0,0)) = 1-\rho^-,\\
&\mathbb{P}(W_{t+1}=1|\boldsymbol{W}_{t-1:t}=(0,1)) = \mathbb{P}(W_{t+1}=0|\boldsymbol{W}_{t-1:t}=(1,0)) = \rho^+ \ge \rho,\\
&\mathbb{P}(W_{t+1}=0|\boldsymbol{W}_{t-1:t}=(0,1)) = \mathbb{P}(W_{t+1}=1|\boldsymbol{W}_{t-1:t}=(1,0)) = 1-\rho^+.
\end{aligned} \tag{21}$$

where $\rho^-\rho + \rho^+(1-\rho) = \rho$. Once we set ρ and ρ^+, ρ^- is also known, so only introduce one more parameter in this extension.

Intuitively, this refined design not only balances the observations with respect to seasonality, but further couples the observations temporally. The latter means that within any short time window (e.g., 8 periods), the numbers of treatment and control observations do not differ significantly. Particularly, for $\rho = 0.5, \rho^- = 0, \rho^+ = 1$, we will end up with a design that generates the assignment path with perfect temporal coupling for each day (i.e., $\boldsymbol{w}_{t-7:t} = (1,1,0,0,1,1,0,0)$). This is typically known as mirror strategy in practice, an extension of match-pair design (Bai 2019) in the interference setting.

To study the theoretical property of this refined design, we note that due to the constraint $\rho^-\rho + \rho^+(1-\rho) = \rho$, we maintain the same marginal distribution at each block.

THEOREM 4. *Under the design $\eta^\dagger$, for block $b > 1$, the marginal distribution of the treatment assignments $\boldsymbol{W}^B_{b-1,b}$ still follows a random permutation over the set*

$$S = \left\{ \underbrace{(1,1),\ldots,(1,1)}_{K_v/2}, \underbrace{(0,0),\ldots,(0,0)}_{K_v/2}, \underbrace{(1,0),\ldots,(1,0)}_{(K-K_v)/2}, \underbrace{(0,1),\ldots,(0,1)}_{(K-K_v)/2} \right\}.$$

The proof is given in Appendix H.

Algorithm 2 New Design $\eta^\dagger$ with Temporal Coupling

1: Input: K, $2 \le K_v \le K(\rho = K_v/K)$, $\rho^- \le \rho$ and $\rho^+ \ge \rho$ with $\rho^-\rho + \rho^+(1-\rho) = \rho$.
2: Generate the assignments of the first block $\boldsymbol{W}_1^B$ following a complete randomization with $K/2$ treatments and $K/2$ controls.
3: **for** $b \in \{2, \ldots, B\}$ **do**
4: Let $\mathcal{T}^1$ be the set of periods where for $t \in \mathcal{T}^1$, there exists some k such that $t = (b-1)+(k-1)B$ and $W_t = 1$; Let $\mathcal{T}^0$ be the set of periods where for $t \in \mathcal{T}^0$, there exists some k such that $t = (b-1)+(k-1)B$ and $W_t = 0$.
5: To generate the assignments $W_{t+1}, \forall t \in \mathcal{T}^1$, we create a set that contains $K_v/2$ treatments and $(K-K_v)/2$ controls.
6: **if** $b = 2$ **then**
7: We randomize with $\mathbb{P}(W_{t+1} = 1|W_t = 1) = \rho$ and $\mathbb{P}(W_{t+1} = 0|W_t = 1) = 1-\rho$.
8: **else**
9: We randomize with $\mathbb{P}(W_{t+1} = 1|\boldsymbol{W}_{t-1:t} = (0,1)) = \rho^+$, $\mathbb{P}(W_{t+1} = 0|\boldsymbol{W}_{t-1:t} = (0,1)) = 1-\rho^+$, $\mathbb{P}(W_{t+1} = 1|\boldsymbol{W}_{t-1:t} = (1,1)) = \rho^-$, and $\mathbb{P}(W_{t+1} = 0|\boldsymbol{W}_{t-1:t} = (1,1)) = 1-\rho^-$.
10: **end if**
11: We symmetrize Step 5-10 for $W_{t+1}, \forall t \in \mathcal{T}^0$.
12: **end for**
13: Return the assignment path $\boldsymbol{W}_T$.

5.2. Analysis of Bias and Variance

To compare with the original design η, we follow the same bias and variance analysis. First, we show that the bias upper bound can be adjusted using the design parameter ρ^-.

COROLLARY 3. *Under Assumption 2, the bias of the refined design η^* in Algorithm 2 with parameters ρ and ρ^- has an upper bound:*

$$|\tau(\mathbb{Y}) - \bar{\tau}(\mathbb{Y}, \eta)| \le \frac{1}{T}\sum_{t=1}^{T} 2\rho^-\phi_t(3) + 2(1-\rho^-)\phi_t(2). \tag{22}$$

We omit the proof as it simply modifies the proof of Theorem 2 by changing ρ to ρ^-. Since $\rho^- \le \rho$, the bias gets larger in the refined design. The intuition is that the refined design reduces the appearance of longer consecutive treatment assignments such as $\boldsymbol{w}_{t-5:t} = \mathbf{1}_6$, and enhances the chance of observing sequences like $\boldsymbol{w}_{t-5:t} = (1,1,0,0,1,1)$.

Next, because the refined design preserves the same property of the marginal distribution at each block, its variance can be decomposed in the same way as Theorem 3. To accommodate the refined

design, we extend the definition of overlapping indices in (13) and define

$$\begin{aligned}\gamma^{+}_{t,t'}(\rho,\rho^{+}) &= \frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1)|\boldsymbol{W}_{t-1:t}=(1,1))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1))}\\ \gamma^{-}_{t,t'}(\rho,\rho^{+}) &= \frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0)|\boldsymbol{W}_{t-1:t}=(1,1))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0))}.\end{aligned} \tag{23}$$

The key difference compared to the original design η is on the numerators since the refined design η^* directly adjusts the conditional probability through ρ^+. Similar to Theorem 3, there are no closed-form expressions for $\gamma^{+}_{t,t'}(\rho,\rho^{+})$ and $\gamma^{-}_{t,t'}(\rho,\rho^{+})$, and we derive their values through recursion. Here, we focus on the discussion of the variance upper bound.

COROLLARY 4. *Under Assumption 2 - 3, the variance of the refined design η^* in Algorithm 2 with parameters ρ and ρ^+ can be decomposed into two parts where*

$$\begin{aligned}V_1 =& \frac{1}{B^2K}\sum_{b=1}^{B}(\frac{2}{\rho}(S_b^1+S_b^0)-S_b^\tau)\\ &+\frac{1}{B^2K}\sum_{b\neq b'\in[B]}((\gamma^{+}_{b,b'}(\rho,\rho^{+})-\gamma^{-}_{b,b'}(\rho,\rho^{+}))(S_{b,b'}^{1,1}+S_{b,b'}^{0,0})-(1-\gamma^{-}_{b,b'}(\rho,\rho^{+}))S_{b,b'}^{\tau}).\\ V_2 \leq& V_2^{\mathrm{U}} = \frac{4\rho^-(1-\rho^-)}{B}\left(\frac{\phi(2)^2}{\rho(K-1)}+\frac{\sum_{q=3}^{\infty}\phi(q)^2}{\rho K/2-1}\right)\\ &+\frac{4\rho^-(1-\rho^-)}{B^2}\sum_{b\neq b'\in[B]}\left(\left(\frac{1}{K}+\frac{1}{\rho K-2}\right)\phi(2)\phi(2+\delta_{b,b'})+\left(\frac{1}{K}+\frac{1}{\rho K/2-1}\right)\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'})\right).\end{aligned} \tag{24}$$

The proof is given in Appendix I. The variance of the refined design follows the same structure as Corollary 1 - 2, and only differs in the weights, which are discussed below. For the first part V_1, the weights $\gamma^{+}_{b,b'}(\rho,\rho^{+})-\gamma^{-}_{b,b'}(\rho,\rho^{+})$ and $(1-\gamma^{-}_{b,b'}(\rho,\rho^{+})$ will reduce to the ones in Corollary 2 when $\rho^+=\rho$, but in general, there are no closed-form expressions. To get more intuition for the comparison of this part between the refined design $\eta^\dagger$ and the original one η, we present the actual values of the weights under certain parameter configurations in Figure 4. As we discussed before, the magnitude of $S_{b,b'}^{1,1}+S_{b,b'}^{0,0}$ is typically much larger than $S_{b,b'}^{\tau}$ so we particularly show how the weight $\gamma^{+}_{b,b'}(\rho,\rho^{+})-\gamma^{-}_{b,b'}(\rho,\rho^{+})$ changes in $\delta_{b,b'}$. This weight determines how much the covariance of the outcomes from two block b and b' contributes to the total variance. The figure illustrates the weight distribution under four different parameter configurations: $(\rho=1,\rho^+=1)$, $(\rho=3/4,\rho^+=3/4)$, $(\rho=1/2,\rho^+=1/2)$, and $(\rho=3/4,\rho^+=7/8)$. The first three configurations correspond to our original design, while the fourth introduces a refinement. The comparison reveals a clear trade-off in weight allocation between smaller and larger distances $\delta_{b,b'}$. When $\rho=\rho^+=1$, the design collects the maximum number of samples, but at the expense of assigning significant weights to the covariance between distant blocks. In contrast, when $\rho=\rho^+=0.5$, the design completely

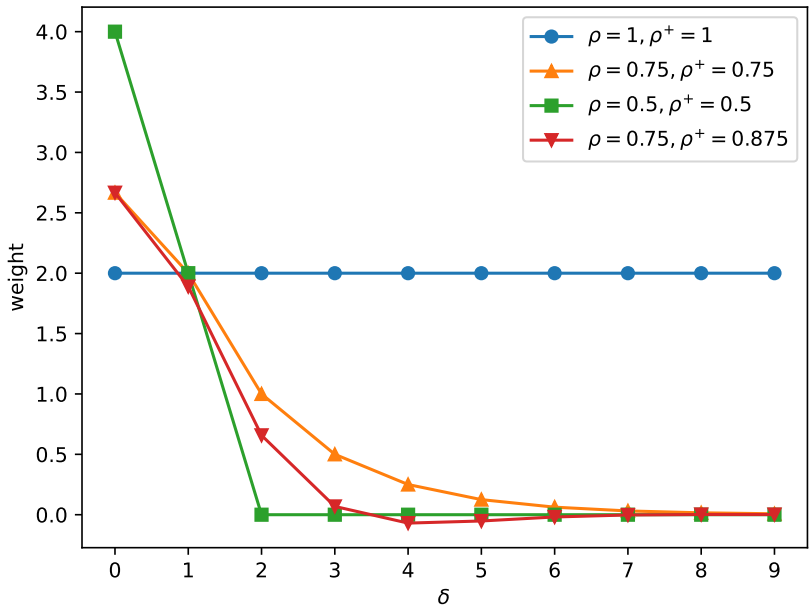


Figure 4 **The value of weight $\gamma^+_{b,b'}(\rho,\rho^+) - \gamma^-_{b,b'}(\rho,\rho^+)$ for the finite sample covariance of outcomes $S^{1,1}_{b,b'} + S^{0,0}_{b,b'}$ from two blocks with distance $\delta_{b,b'}$.**

ignores the weights for $\delta_{b,b'} \geq 2$, which leads to the highest variance contribution within individual blocks. The configuration $\rho = \rho^+ = 0.75$ strikes a balance between these two extremes. In particular, setting $\rho^+ = 0.875$, the weight curve is shifted downward, resulting in generally lower weights across distances. This adjustment is beneficial because it reduces the overall variance, except in extreme cases where there is a strong negative correlation between blocks with $\delta_{b,b'} \geq 4$. Consequently, the refined design is expected to exhibit a lower V_1 under most practical conditions, thereby enhancing the efficiency and robustness of the experiment. For the second part V_2^{U}, as we discussed in Assumption 2, $\phi_{t+\delta_{b,b'}}(q+\delta_{b,b'})$ is typically much smaller than $\phi_t(q)$ for $\delta_{b,b'} \geq 1$, so the first row of V_2^{U} primarily determines the upper bound. The main distinction compared to Corollary 1 is replacing $\rho(1-\rho)$ with $\rho^-(1-\rho^-)$. Setting ρ^- close to 0 (that is, ρ^+ close to 1) results in a high probability of observing $\boldsymbol{W}_{t-2} = 0$ conditioning on $\boldsymbol{W}_{t-1:t} = (1,1)$, reducing the variation from distant assignments.

Based on this discussion, we emphasize that the refined design lays the groundwork for numerous potential extensions inspired by the concept of temporal coupling, where assignments in each block may depend on a more extensive history of past blocks. For instance, rather than simply increasing the conditional probability of generating $W_t = 0$ after observing two consecutive treatments $\boldsymbol{W}_{t-2:t-1} = (1,1)$, it may be beneficial to consider increasing this probability only after three consecutive treatments $\boldsymbol{W}_{t-3:t-1} = (1,1,1)$. Such an approach could yield more observations and improve bias reduction. However, exploring these more intricate extensions is beyond the scope of

this paper, as our current focus is on demonstrating how temporal coupling within the model-assisted design framework for switchback experiments can improve the original design.

6. Inference and Testing

This section explains how to conduct hypothesis tests for the treatment effect using the data collected from a switchback experiment following our design.

We first introduce two notations. Let $m_b^{u,v} = \max_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},u,v)] - \overline{Y_b(\cdot,u,v)})^2$ and $r_b^{u,v} = \frac{1}{K-1}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},u,v)] - \overline{Y_b(\cdot,u,v)})^2$. Under the assumption that as $T\to\infty$, if $\max_{u\in\{0,1\},v\in\{0,1\}} m_b^{u,v}/(T\cdot r_b^{u,v}) \to 0$ for all $b\in[B]$, then we have the finite population central limit theorem for the estimator (2). Informally, this requirement guarantees that the variance of outcomes within each block is not dominated by any single period; rather, variability is spread thinly across the growing time horizon. Such "no-single-observation dominance" conditions mirror the standard criterion in classical central-limit-theorem proofs and are routinely imposed in large-sample analyses of experiments.

THEOREM 5. *The limiting distribution of $\widehat{\tau}$ using our new design η is a normal distribution, i.e., as $T\to\infty$,*

$$\frac{\widehat{\tau}-\mathbb{E}[\widehat{\tau}]}{\sqrt{\mathrm{Var}_\eta(\widehat{\tau})}} \xrightarrow{D} \mathcal{N}(0,1). \tag{25}$$

This gives the nominal coverage that $\lim_{T\to\infty}\mathbb{P}(|\widehat{\tau}-\mathbb{E}[\widehat{\tau}]|/\sqrt{\mathrm{Var}_\eta(\widehat{\tau})}\geq c(\alpha)) = \alpha$, where α is the nominal level and $c(\alpha)$ is the critical value of the standard normal distribution.

We leave the proof in Appendix F. Note that the CLT is centered around $\mathbb{E}[\widehat{\tau}]\neq\tau$, indicating that the estimator (2) is asymptotically biased. This arises because, while the variance of $\widehat{\tau}$ diminishes as T grows, the bias remains fixed—since the estimator itself does not adapt to T. To ensure consistency (i.e., $\widehat{\tau}\to\tau$ in probability), one must adapt the estimator to the horizon length T, which in turn requires assumptions on the underlying system dynamics.

Since the true variance of the estimator is unknown, to apply the CLT for inference, we need to derive a variance estimator. This can be accomplished by constructing a series of sample statistics $R_{b,b'}^{u,v}, \forall u\in\{0,1\}, v\in\{0,1\}$. For ease of exposition, we take $b=b', u=v$ as an example. Given a realized assignment path $\boldsymbol{w}_{1:T}^{\mathrm{obs}}$, we collect the valid observations in a block and define the sample mean

$$\widehat{Y_b(\cdot,u,u)} = \frac{1}{K_v/2}\sum_{t\in\mathcal{T}_b} Y_t(\boldsymbol{w}_{1:t-2}^{\mathrm{obs}},u,u)\mathbb{1}\{\boldsymbol{w}_{t-1:t}^{\mathrm{obs}}=(u,u)\} \tag{26}$$

which further gives rise to

$$R_b^u \equiv R_{b,b}^{u,u} = \frac{1}{K_v/2-1}\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{w}_{1:t-2}^{\mathrm{obs}},u,u) - \widehat{Y_b(\cdot,u,u)})^2\mathbb{1}\{\boldsymbol{w}_{t-1:t}^{\mathrm{obs}}=(u,u)\}. \tag{27}$$

We derive other $R^{u,v}_{b,b'}$ in the same way, and construct the following variance estimator:

$$\widehat{\mathrm{Var}_\eta}(\widehat{\tau}) = \frac{1}{B^2K}\sum_{b,b'\in[B]}(\gamma^+_{b,b'}(R^{1,1}_{b,b'}+R^{0,0}_{b,b'}) - \gamma^-_{b,b'}(R^{1,0}_{b,b'}+R^{0,1}_{b,b'})). \tag{28}$$

Proposition 1. *The variance estimator satisfies*

$$\begin{aligned}
\mathbb{E}[\widehat{\mathrm{Var}_\eta}(\widehat{\tau})] &= \mathrm{Var}_\eta(\widehat{\tau}) + \frac{1}{K(K-1)}\sum_{k=1}^{K}(\bar{\tau}_k-\bar{\tau})^2 \\
&\quad - \frac{1}{B^2K(K-1)}\sum_{b,b'\in[B]}\sum_{\substack{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}\\ |t'-t|\neq\delta_{b,b'}}}(\sigma^{1,1}_{t,t'}+\sigma^{0,0}_{t,t'}-\sigma^{1,0}_{t,t'}-\sigma^{0,1}_{t,t'}) \\
&\geq \mathrm{Var}_\eta(\widehat{\tau}) - \frac{2\rho(1-\rho)}{B(K-1)(K/2-1)}\left(\phi(2)^2 + K\sum_{q=3}^{\infty}\phi(q)^2\right) \\
&\quad - \frac{2\rho(1-\rho)}{B^2(K/2-1)}\sum_{b\neq b'\in[B]}\left(\phi(2)\phi(2+\delta_{b,b'}) + \sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'})\right).
\end{aligned} \tag{29}$$

We leave the details of the sample statistics and the proof in Appendix G. The key property is that the expectation of the sample statistics $R^{u,v}_{b,b'}$ contains the covariance from $S^{u,v}_{b,b'}$ and the part of the conditional covariance from $\sigma^{u,v}_{t,t'}$. As a result, it leaves with two residuals. The first one involving the variance of treatment effect is common in the study of complete randomization. Though not identifiable, it is often ignored due to the non-negativity. The second one involving the conditional covariance $\sigma^{u,v}_{t,t'} = \mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},u,u), Y_{t'}(\boldsymbol{W}_{1:t'-2},v,v))$ is not identifiable as well because we have at most one sample of $Y_t(\boldsymbol{w}^{\mathrm{obs}}_{1:t-2},u,u)$ and $Y_{t'}(\boldsymbol{w}^{\mathrm{obs}}_{1:t'-2},v,v)$. Unfortunately, the sign is indeterminate, so we leverage the analysis in Corollary 1 to obtain the inequality in (29). Note that the true variance $\mathrm{Var}_\eta(\widehat{\tau})$ is on the order of $\frac{\phi(2)^2}{K}$. For the remainder, $\frac{2\rho(1-\rho)}{B(K-1)(K/2-1)}\phi(2)^2$ decreases faster as K grows while the other three terms would be small if $\phi(q)$ decays faster in q (e.g., diminishes exponentially under the geometric mixing condition), so the expectation of the variance estimator approximately covers the true variance. In conjunction with the bias analysis, this implies that, in practice, a properly defined truncated estimator (the duration of each period) is helpful—not only to reduce bias, but also to have an accurate variance estimate.

7. Numerical Study

In this section, we conduct a comprehensive numerical study to investigate the performance of our new design, as well as its practical implementations. This numerical study is two-fold. First, we assume that the outcome model follows a time series model fitted from the trip data Taxi and Commission (2025). We then specify the form of treatment effects and adjust other parameters to compare various approaches. Second, we consider a real-world setting in which all data is generated by a ride-sharing simulator (Farias et al. 2022, Hu and Wager 2022) based on the same trip data Taxi and Commission (2025). In this case, the outcome model is very complicated and implicit, so it provides a solid robustness check for practice.

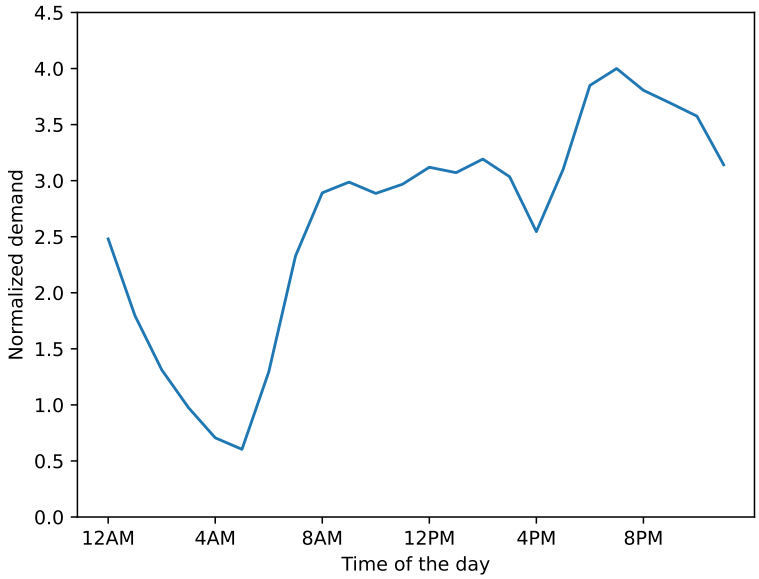


Figure 5 **Normalized demand distribution in NYC taxi trip data**

7.1. Time Series Models

We begin by specifying the Bayesian structural time series (4) as the data generation process for the simulation:

$$\begin{aligned} Y_t(\boldsymbol{w}_{1:t}) &= \alpha_t + s_t + \lambda_t^Y w_t + \varepsilon_t, \\ \alpha_t &= \gamma\alpha_{t-1} + \lambda_t^\alpha w_t + \beta_t + \eta_t, \\ \beta_t &= \theta\beta_{t-1} + \xi_t. \end{aligned} \tag{30}$$

where the residuals follow Gaussian with unknown variances ($\varepsilon_t \sim N(0,\sigma_\epsilon)$, $\eta_t \sim N(0,\sigma_\eta)$ and $\xi_t \sim N(0,\sigma_\xi)$). We fit the model without intervention($\boldsymbol{w}$) using the normalized NYC taxi trip data Taxi and Commission (2025) from Jan, 2013 to Apr, 2013, where the gap between two timestamps t and $t+1$ is set to be 30 minutes and $Y_t(\boldsymbol{w}_{1:t}) \in [0,4]$. [6] First, the seasonality component s_t with period $B = 48$ is the most significant, as shown by Figure 2 in Section 1. Figure 5 summarizes the fitted daily pattern with a valley during the early morning and a peak during the evening. We show a two-week forecast from the fitted model in Figure 6. We also present the estimated parameters in Table 1 and use them as the "true" model to generate new data in this simulation.

We consider the following specific choices of other parameters. We specify a constant direct and indirect effect $\lambda_t^Y = 0.4$, $\lambda_t^\alpha = -0.15$. Under this setting, we can derive the true global average treatment effect $\tau = \lambda^Y + \lambda^\alpha/(1-\gamma) \approx -0.1$, while the naive difference-in-means estimator gives

[6] This setup is flexible that for the design, we can easily merge time intervals to define a period with duration 30, 60 or 90 minutes.

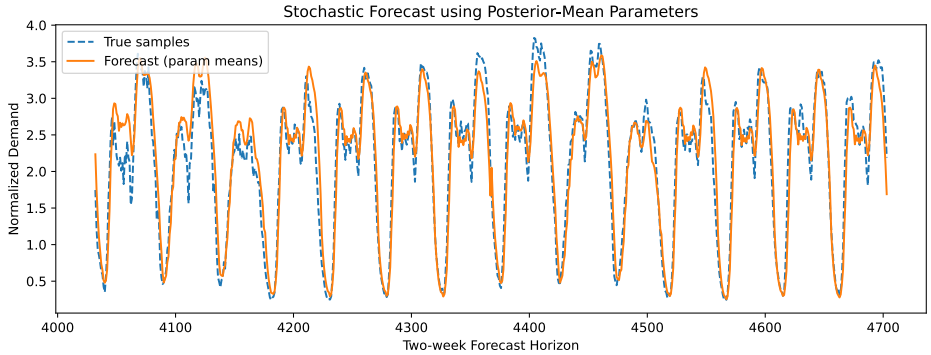


Figure 6 **Two-week forecast from the fitted model**

parameter	γ	θ	σ_η	σ_ξ	σ_ϵ
mean	0.702	0.622	0.103	0.156	0.035
sd	0.022	0.027	0.003	0.002	0.002

Table 1 **Estimated parameters in the time series**

$\hat{\tau} = \lambda_t^Y - \lambda_t^\alpha = 0.25$. That is, the bias (equal to 0.35) incurred by ignoring the carryover effects leads to the opposite conclusion.

We will compare four classes of methods, differentiating in the design and the estimator.

- BN(l): the optimal design (Bojinov et al. 2022) of Bernoulli experiments with IPW estimator by assuming limited carryover effects, parameterized by the period length l. That is, $l = 2$ means generating the assignment every 2×30 mins.
- BN(b, l): the design (Hu and Wager 2022) of Bernoulli experiments with burn-in and bias-correction estimator by assuming geometric mixing condition, parameterized by both the burn-in window b and the switching window l. Typically, we have $l > b$ so the estimator only discards part of the data that is close to the switching point. Once we let $l = b$, it actually reduces to the first method.
- CR(l): the design (Ni et al. 2023) of completely randomized experiments with IPW estimator is another extension of BN(l), where the number of treatment and control observations are guaranteed to be equal, but still random.
- MA(l, ρ): the model-assisted design η in this paper, parameterized by the period length l and the design parameter ρ.
- MA†(l, ρ, ρ^+): the variant using temporal coupling η^* with the additional design parameter ρ^+.

For each method, we consider several parameter configurations and analyze bias, standard error, and RMSE.

The time horizon of the experiment is set to be two weeks ($K = 14$). We first summarize the results of three benchmarks in Table 2. It is not surprising that BN(l) performs the worst due to the

Design & Estimator	Bias	Standard Error	RMSE
BN($l=1$)	0.245	0.368	0.443
BN($l=2$)	0.151	0.511	0.533
BN($l=3$)	0.087	0.625	0.631
BN($b=1,l=3$)	0.201	**0.140**	0.245
BN($b=1,l=4$)	0.175	0.156	0.235
BN($b=2,l=4$)	0.133	0.181	0.225
BN($b=2,l=6$)	0.101	0.200	**0.224**
BN($b=3,l=6$)	**0.076**	0.213	0.226
CR($l=1$)	0.243	0.158	0.290
CR($l=2$)	0.147	0.220	0.265
CR($l=3$)	0.085	0.259	0.273

Table 2 The performance of benchmark methods

Design & Estimator	Bias	Standard Error	RMSE
MA($l=1,\rho=6/7$)	0.144	0.118	0.186
MA($l=1,\rho=5/7$)	0.199	**0.108**	0.227
MA($l=2,\rho=6/7$)	**0.053**	0.136	**0.146**
MA($l=2,\rho=5/7$)	0.078	0.134	0.155
MA($l=2,\rho=4/7$)	0.098	0.136	0.167
MA†($l=2,\rho=6/7,\rho^+=13/14$)	0.056	0.133	**0.144**
MA†($l=2,\rho=5/7,\rho^+=6/7$)	0.093	0.121	0.153
MA†($l=2,\rho=4/7,\rho^+=11/14$)	0.121	0.108	0.161

Table 3 The performance of the new designs

large standard errors. In comparison, as suggested in Hu and Wager (2022), BN(b,l) with a judicious use of burn-in periods greatly improves the standard errors. Similar to the numerical results in Hu and Wager (2022), we also find that RMSE is relatively stable under different choices of b and l. Moreover, CR(l) is also efficient in reducing the standard errors. This phenomenon is also discussed in Ni et al. (2023), highlighting the importance of balancing in the design of switchback experiments. That said, this simple complete randomization still performs worse than BN(b,l).

We then pivot to Table 3 for the performance of the new design. Given that $K=14$, we consider three choices of design parameters $\rho\in\{4/7,5/7,6/7\}$. We observe that the standard error and RMSE are consistently smaller than the simple complete randomization CR(l) and BN(b,l). Specifically, comparing BN$(b=3,l=6)$ and MA$(l=2,\rho=5/7)$, we notice that they share the almost same bias (0.076/0.078), but the standard error is greatly reduced ($0.213\to0.134$). Consistent with our theoretical analysis, we see that the bias is decreasing in ρ while the standard error is not necessarily monotone in ρ. Both $\rho=5/7$ and $\rho=6/7$ achieve the superior performance, hitting a better balance of bias-variance tradeoff than the benchmarks. We also present the performance of the refined design with temporal coupling. We simply consider $\rho^+=0.5(1+\rho)$ in the simulation. As anticipated from our analysis, the results indicate that MA† exhibits a slightly smaller variance but a larger

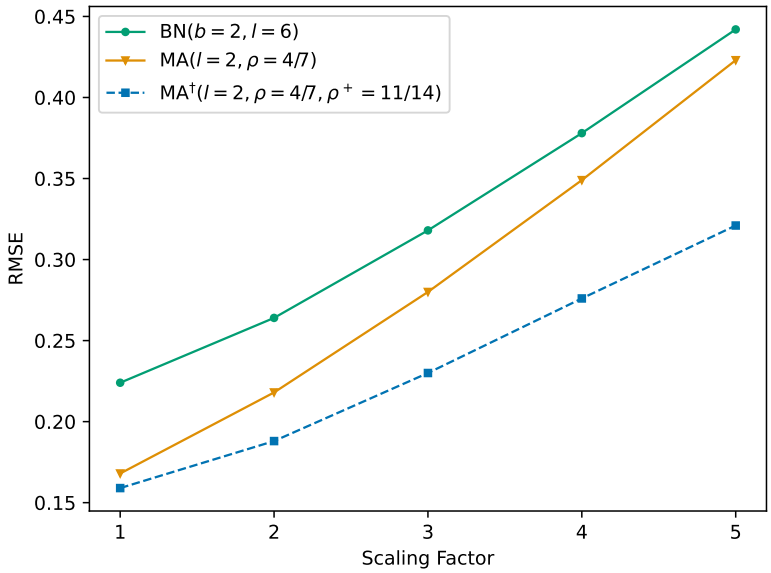


Figure 7 **Scaling the residual variances**

bias compared to MA, achieving a similar RMSE overall and underscoring its potential to improve estimation accuracy in practice.

We also examine the practical strategy of choosing ρ based on the first-order condition (20):

$$-\sum_{b=1}^{B}\frac{\widehat{S}_b}{\rho^2}+\sum_{b\neq b'\in[B],\delta_{b,b'}\geq 2}2(\delta_{b,b'}-1)(2\rho-1)^{\delta_{b,b'}-2}\widehat{S}_{b,b'}=0. \tag{31}$$

Because the weight of $\widehat{S}_{b,b'}$ shrinks exponentially as $\delta_{b,b'}$ increases, we only keep the terms up to $\delta_{b,b'}\leq 4$. Taking $l=2$ as an instance where we have $B=24$ blocks, we estimate $\widehat{S}_b$ and $\widehat{S}_{b,b'}$ using the same trip data mentioned above. The line search for the first-order condition leads to $\hat{\rho}\approx 0.67$ that minimizes the first part of the variance (18). Following the rule that $\rho\in(0.67,1]$, we eliminate $\rho=4/7$ and only consider $\rho=5/7$ or $\rho=6/7$ depending on our tolerance to the bias. This is consistent with the empirical performance in Table 2.

Next, we introduce several adjustments to the simulation model (30) to better evaluate the performance of our model-assisted design under varying conditions. First, we observe from Table 2 that the performances of MA and MA† are quite similar. This is consistent with the fitted time-series model, where residual variances are small and seasonality explains most of the variation in outcomes. In such settings, the primary benefit of our method comes from achieving strong balance between treatment and control observations within blocks, with only minor gains from additional temporal coupling. However, this favorable scenario may not hold for other outcomes of interest across all business contexts. In many applications, the observed variation in outcomes cannot be fully explained

by seasonality alone. To quantify this, note that under a non-experimental environment, the variance of the observed outcome is given by:

$$\mathrm{Var}\left(y_t\right) = \mathrm{Var}\left(s_t\right) + \sigma_\varepsilon^2 + \frac{\sigma_\eta^2\left(1-\theta^2\right)\left(1-\gamma\theta\right) + \sigma_\xi^2(1+\gamma(2-\theta))}{\left(1-\gamma^2\right)\left(1-\theta^2\right)\left(1-\gamma\theta\right)}. \tag{32}$$

To explore how increased temporal noise affects design performance, we gradually scale the fitted values of σ_η and σ_ξ from Table 1 and track the resulting changes in RMSE in Figure 7. As the scale factor increases, the performance gap between BN and MA narrows, indicating diminishing returns from block-level balancing. In contrast, MA†—our refined design with temporal coupling—maintains stable performance even under high variance. This suggests a "doubly robust" property: the refined new design adapts well whether seasonality or auto-correlation is the dominant feature in the time series.

Up to this point, our simulations were based on the time-series model (30). We now introduce further complexity to the data-generating process by considering the following extension:

$$\begin{aligned} Y_t(\boldsymbol{w}_{1:t}) &= \alpha_t + s_t(1+\lambda_t^Y w_t) + \varepsilon_t, \\ \alpha_t &= (\gamma + \lambda_t^\gamma w_t)\alpha_{t-1} + \beta_t + \lambda_t^\alpha w_t + \eta_t, \\ \beta_t &= \theta\beta_{t-1} + \theta'\beta_{t-2} + \xi_t \end{aligned} \tag{33}$$

where s_t, γ, θ, λ_t^α and three residuals remain the same as before. We let $\lambda_t^Y = 0.1$, $\lambda_t^\gamma = -0.05$ and $\theta' = 0.2$. This revised model introduces three layers of complexity:

- The local trend β_t follows a second-order autoregressive process, increasing temporal dependence.
- The direct treatment effect is now multiplicative with the seasonal component s_t, rather than additive.
- The intervention modifies the system's dynamics by altering the decay rate γ, introducing a form of non-stationarity in response to treatment.

These changes present significant challenges for model-based approaches, where accurate estimation requires a precise specification of how treatment interacts with the system. In contrast, our model-assisted design remains effective, as it does not rely on estimating or plugging in any model coefficients for inference. As shown in Table 4, our designs continue to perform well despite the increased complexity. Notably, the performance gap between MA and MA† widens relative to Table 3, reflecting the stronger auto-correlation in this setting and the additional value of temporal coupling. We may again follow the practical strategy of choosing ρ by checking the first-order condition (20), though the treatment will impact the covariance between periods in (33). The line search outputs $\hat{\rho} \approx 0.53$, which is much smaller than the one under the model (30). This suggests $\rho \in (0.53, 1]$ and thus $\rho = 4/7$ or $\rho = 5/7$ should be good choices, as we see in Table 4.

Design & Estimator	Bias	Standard Error	RMSE
BN($b=1,l=4$)	0.137	0.168	0.233
BN($b=2,l=4$)	0.102	0.188	0.234
BN($b=2,l=6$)	0.078	0.209	0.245
MA($l=2,\rho=6/7$)	**0.054**	0.193	0.201
MA($l=2,\rho=5/7$)	0.081	0.177	0.195
MA($l=2,\rho=4/7$)	0.105	0.171	0.201
MA†($l=2,\rho=6/7,\rho^+=13/14$)	0.057	0.184	0.193
MA†($l=2,\rho=5/7,\rho^+=6/7$)	0.098	0.154	**0.183**
MA†($l=2,\rho=4/7,\rho^+=11/14$)	0.133	**0.127**	0.184

Table 4 The performance of the new designs under the complex time series

7.2. Ride-sharing Examples

We conduct another more practical example used in Hu and Wager (2022) based on a comprehensive ride-sharing simulator in Farias et al. (2022). This simulator models driver and rider dynamics using data derived from the NYC taxi trip records dataset. Drivers enter the system continuously, each with the capacity to transport up to three riders. Their initial locations are sampled randomly from the dataset, and their shifts follow an exponential duration model. Drivers exit the system once their shifts conclude. Riders make requests at random times, with pick-up and drop-off points selected from the trip records dataset. Their value-for-time parameter, which influences their decision to accept an offer, is drawn from a lognormal distribution. When a request is initiated, the system dispatches a driver based on the current policy. The rider receives an offer reflecting the expected cost of service from the assigned driver and decides whether to accept. If accepted, the driver's route is updated to include the new rider.

Similar to Hu and Wager (2022), we explore two dispatching policies that differ in the frequency of allocating a pool driver. Specifically, a pool driver is selected if the additional travel cost is less than κ times that of an idle driver. We set $\kappa=1$ as the control policy and $\kappa=0.5$ as the treatment policy. The outcome of interest is the aggregated profit, calculated as the total revenue collected from riders minus the operational costs incurred by the ride-sharing company for completed trips. Changes to the dispatch policy can have carryover effects on future outcomes by affecting the system dynamics: for instance, assigning a driver to a request affects their subsequent availability and location, thereby influencing future requests and assignments. These cascading effects can persist over time and extend to nearby drivers. As illustrated in Hu and Wager (2022), such effects diminish fast and the system's state gradually stabilizes, although full convergence is not achieved.

In this paper, we mostly follow the same parameter configurations as in Hu and Wager (2022), such as the pricing policy, the cost function, the utility models, etc, but adjust the design parameters. For example, in Hu and Wager (2022), the burn-in length is chosen to be 800 or 1600 seconds, and the switching length is 4000 seconds. Since our new design leverages the seasonality, we adjust the minimal

Design & Estimator	Bias	Standard Error	RMSE
CR($l=1$)	0.432	2.02	2.06
CR($l=2$)	0.275	2.74	2.75
BN($b=1,l=2$)	0.423	1.70	1.75
BN($b=1,l=3$)	0.307	1.85	1.87
BN($b=2,l=4$)	0.265	2.30	2.32
BN($b=2,l=6$)	0.201	2.51	2.52
MA($l=1,\rho=6/7$)	**0.130**	1.80	1.81
MA($l=1,\rho=5/7$)	0.224	1.60	1.62
MA($l=1,\rho=4/7$)	0.331	1.68	1.71
MA†($l=1,\rho=5/7,\rho^+=6/7$)	0.363	1.45	1.49
MA†($l=1,\rho=4/7,\rho^+=11/14$)	0.427	**1.18**	**1.25**

Table 5 The performance of different methods in the ridesharing example

length of each period to be 30 minutes (i.e. 1800 seconds). We consider an experimental horizon with two weeks. Since BN(l) is strictly dominated, we only compare the other three approaches in Table 5. It is not surprising to see a bias–variance trade-off across different parameter configurations. In particular, BN(b,l) is better than CR(l) by achieving lower variance for similar levels of bias. MA(l,ρ) further improves this balance compared to the two benchmarks, and is further surpassed by MA$^\dagger(l,\rho,\rho^+)$, due to a notable reduction in variance. This indicates a non-negligible variation in daily outcomes and underscores the benefit of leveraging temporal coupling to achieve more accurate estimates.

The ride-sharing example confirms that our new designs, especially the refined one, exhibit consistently superior performance than existing methods. This robustness suggests that it can be trusted to maintain good performance across different settings.

8. Concluding Remarks

In this study, we propose a novel model-assisted design of switchback experiments, tackling the challenges in practical implementations. By incorporating seasonality balancing and adaptive assignment strategy, our new design improves both the efficiency and stability of experimental results, facilitating the wide adoption of switchback experiments. The numerical study with both simple time series models and real-world ridesharing simulation demonstrates the superior performance of the new design, entailing a better bias-variance tradeoff compared to existing solutions.

This study primarily focuses on the design aspect, with a brief exploration of the estimation part. However, it is increasingly popular to integrate machine learning techniques in causal inference (Chernozhukov et al. 2024), making it an intriguing area for future research. Exploring the impact of such techniques on switchback experiments could provide valuable insights and potentially enhance their efficiency. Additionally, expanding the application of our design to other contexts, such as different experimental settings, could provide further validation and refinement of the methodology.

Overall, this research contributes a valuable framework for conducting rigorous and efficient switchback experiments in environments where traditional A/B testing is not feasible or reliable, paving the way for more informed and effective decision-making in operations.

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Online Appendix

A. Proof of Theorem 1

We prove the theorem using induction. We first check the block $b=2$. Based on our randomization mechanism, it is guaranteed that we observe $K_v/2$ of $(1,1)$, $K_v/2$ of $(0,0)$, $(K-K_v)/2$ of $(1,0)$, $(K-K_v)/2$ of $(0,1)$. There are $Q=\frac{K!}{(K_v/2)!(K_v/2)!((K-K_v)/2)!((K-K_v)/2)!}$ number of feasible arrangements $\boldsymbol{w}_2^B$ in the block, and it suffices to show that all of them share the same probability $1/Q$. W.L.O.G., let us consider the one

$$\boldsymbol{w}_{1:2}^B=[(1,1),\ldots,(1,0),\ldots,(0,1),\ldots,(0,0),\ldots]. \tag{34}$$

Because the first block is generated at purely random, we have $\mathbb{P}(\boldsymbol{W}_1^B=\boldsymbol{w}_1^B)=\frac{(K/2)!(K/2)!}{K!}$. Conditioning on the first block, we also randomly permute $K_v/2$ of 1 and $(K-K_v)/2$ of 0 to generate the first half of the second block. Similarly, we randomly permute $K_v/2$ of 0 and $(K-K_v)/2$ of 1 to generate the second half of the second block. Therefore, we have

$$\mathbb{P}(\boldsymbol{W}_{1:2}^B=\boldsymbol{w}_{1:2}^B)=\mathbb{P}(\boldsymbol{W}_1^B=\boldsymbol{w}_1^B)\cdot\mathbb{P}(\boldsymbol{W}_2^B=\boldsymbol{w}_2^B|\boldsymbol{w}_1^B)=\frac{(K/2)!(K/2)!}{K!}\cdot\left[\frac{(K_v/2)!((K-K_v)/2)!}{(K/2)!}\right]^2=1/Q. \tag{35}$$

Next, suppose that the property is true for some $b\geq 2$, we check the block $b+1$. Because we have a random permutation over $K_v/2$ of $(1,1)$, $K_v/2$ of $(0,0)$, $(K-K_v)/2$ of $(1,0)$, $(K-K_v)/2$ of $(0,1)$ for the block b, it directly implies that the assignments $\boldsymbol{W}_b^B$ is a random permutation over $K/2$ of 1 and $K/2$ of 0. Then following the same argument above, we can show that $\boldsymbol{W}_{b:b+1}^B$ satisfies the property.

B. Proof of Theorem 2

We first recall the bias decomposition:

$$\begin{aligned}\tau(\mathbb{Y})-\bar{\tau}(\mathbb{Y},\eta)=&\frac{1}{T}\sum_{t=1}^{T}\left\{Y_t(\mathbf{1})-\sum_{\boldsymbol{w}_{1:t-2}}Y_t\left(\boldsymbol{w}_{1:t-2},1,1\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t}=(\boldsymbol{w}_{1:t-2},1,1)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}\right\}\\&+\frac{1}{T}\sum_{t=1}^{T}\left\{\sum_{\boldsymbol{w}_{1:t-2}}Y_t\left(\boldsymbol{w}_{1:t-2},0,0\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t}=(\boldsymbol{w}_{1:t-2},0,0)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(0,0)\right)}-Y_t(\mathbf{0})\right\}.\end{aligned} \tag{36}$$

It suffices to analyze the first part since the second part is symmetric. In particular, we have

$$\begin{aligned}
&\sum_{\boldsymbol{w}_{1:t-2}} Y_t\left(\boldsymbol{w}_{1:t-2},1,1\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t}=(\boldsymbol{w}_{1:t-2},1,1)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}\\
=&\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\boldsymbol{w}_{1:t-3},1,1,1\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t}=(\boldsymbol{w}_{1:t-3},1,1,1)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}+Y_t\left(\boldsymbol{w}_{1:t-3},0,1,1\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t}=(\boldsymbol{w}_{1:t-3},0,1,1)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}\right)\\
=&\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\boldsymbol{w}_{1:t-3},1,1,1\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=1\right)\mathbb{P}\left(\boldsymbol{W}_{t-2:t}=(1,1,1)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}\right)\\
&+\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\boldsymbol{w}_{1:t-3},0,1,1\right)\frac{\mathbb{P}\left(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=0\right)\mathbb{P}\left(\boldsymbol{W}_{t-2:t}=(0,1,1)\right)}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}\right)\\
=&\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\boldsymbol{w}_{1:t-3},1,1,1\right)\rho\mathbb{P}(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=1)+Y_t\left(\boldsymbol{w}_{1:t-3},0,1,1\right)(1-\rho)\mathbb{P}(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=0)\right)
\end{aligned}\tag{37}$$

Plug this into (36), for each period, the bias is

$$\begin{aligned}
&\rho\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\mathbf{1}\right)-Y_t\left(\boldsymbol{w}_{1:t-3},\mathbf{1}_3\right)\right)\mathbb{P}(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=1)\\
&+\rho\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\boldsymbol{w}_{1:t-3},\mathbf{0}_3\right)-Y_t\left(\mathbf{0}\right)\right)\mathbb{P}(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=0)\\
&+(1-\rho)\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\mathbf{1}\right)-Y_t\left(\boldsymbol{w}_{1:t-3},0,1,1\right)\right)\mathbb{P}(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=0)\\
&+(1-\rho)\sum_{\boldsymbol{w}_{1:t-3}}\left(Y_t\left(\boldsymbol{w}_{1:t-3},1,0,0\right)-Y_t\left(\mathbf{0}\right)\right)\mathbb{P}(\boldsymbol{W}_{1:t-3}=\boldsymbol{w}_{1:t-3}|W_{t-2}=1).
\end{aligned}\tag{38}$$

Therefore,

$$\begin{aligned}
&|\tau(\mathbb{Y})-\bar{\tau}(\mathbb{Y},\eta)|\\
\leq&\frac{1}{T}\sum_{t=1}^{T}(\rho\max_{\boldsymbol{w}_{1:t-3}}|Y_t(\mathbf{1})-Y_t(\boldsymbol{w}_{1:t-3},1,1,1)|+(1-\rho)\max_{\boldsymbol{w}_{1:t-3}}|Y_t(\mathbf{1})-Y_t(\boldsymbol{w}_{1:t-3},0,1,1)|)\\
+&\frac{1}{T}\sum_{t=1}^{T}(\rho\max_{\boldsymbol{w}_{1:t-3}}|Y_t(\mathbf{0})-Y_t(\boldsymbol{w}_{1:t-3},0,0,0)|+(1-\rho)\max_{\boldsymbol{w}_{1:t-3}}|Y_t(\mathbf{0})-Y_t(\boldsymbol{w}_{1:t-3},1,0,0)|)\\
=&\frac{1}{T}\sum_{t=1}^{T}2\rho\phi_t(3)+2(1-\rho)\phi_t(2)
\end{aligned}\tag{39}$$

where the last equality follows Assumption 2 about the carryover effect.

C. Proof of Theorem 3

Block Representation and Proof Sketch.

To derive the variance, we write the estimator here:

$$\widehat{\tau}\left(\boldsymbol{W}_{1:T};\mathbb{Y},\eta\right)=\frac{1}{T}\sum_{t=1}^{T}\left\{Y_t\left(\boldsymbol{W}_{1:t-2},1,1\right)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}-Y_t\left(\boldsymbol{W}_{1:t-1},0,0\right)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(0,0)\right)}\right\}.\tag{40}$$

We first rewrite it using a block representation:

$$\widehat{\tau}\left(\boldsymbol{W}_{1:T};\mathbb{Y},\eta\right)=\frac{1}{BK}\sum_{b=1}^{B}\sum_{t\in\mathcal{T}_b}\left\{Y_t\left(\boldsymbol{W}_{1:t-2},1,1\right)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}-Y_t(\boldsymbol{W}_{1:t-2},0,0)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(0,0)\right)}\right\} \tag{41}$$

In this way, the variance of our estimator can be decomposed into the sum of covariances between any two blocks, denoted by $\mathrm{Cov}(\widehat{\tau}_b,\widehat{\tau}_{b'})$. Given the complexity of variance analysis, we sketch the steps of analyzing the covariance $\mathrm{Cov}(\widehat{\tau}_b,\widehat{\tau}_{b'})$:

1. Since each potential outcome can depend on the full history of assignment path $\boldsymbol{W}_{1:t}$ and the estimator only accounts for $\boldsymbol{W}_{t-1:t}$, we leverage this structure to analyze the covariance by conditioning on $\boldsymbol{W}_{t-1:t}$. Based on the law of total variance, the covariance $\mathrm{Cov}(\widehat{\tau}_b,\widehat{\tau}_{b'})$ can be decomposed into (i) the covariance of conditional expectation $\mathrm{Cov}(\mathbb{E}[\widehat{\tau}_b|\boldsymbol{W}^B_{b-1:b},\boldsymbol{W}^B_{b'-1:b'}],\mathbb{E}[\widehat{\tau}_{b'}|\boldsymbol{W}^B_{b-1:b},\boldsymbol{W}^B_{b'-1:b'}])$ and (ii) the expectation of conditional covariance $\mathbb{E}[\mathrm{Cov}(\widehat{\tau}_b,\widehat{\tau}_{b'}|\boldsymbol{W}^B_{b-1:b},\boldsymbol{W}^B_{b'-1:b'})]$.
2. Since the estimator has two separate parts associated with treatment outcomes and control outcomes, respectively, we further decompose the covariance of causal effects into (a) the covariance from treatment outcomes only; (b) the covariance from control outcomes only; (c) the covariance from their interactions.

We basically need to derive the six components and add them up, and finally aggregate all pairs of blocks.

The Case of $b=b'$.

To better illustrate the decomposition, we start with the case when the two blocks are identical, which reduces to the variance within a block $\mathrm{Var}(\widehat{\tau}_b)$. Particularly, we have

$$\mathrm{Var}(\widehat{\tau}_b)=\frac{1}{K^2}\,\mathrm{Var}\left(\sum_{t\in\mathcal{T}_b}\left\{Y_t\left(\boldsymbol{W}_{1:t-2},1,1\right)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)}-Y_t(\boldsymbol{W}_{1:t-2},0,0)\frac{\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}}{\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(0,0)\right)}\right\}\right). \tag{42}$$

From the randomization mechanism, we have $\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(0,0)\right)=\mathbb{P}\left(\boldsymbol{W}_{t-1:t}=(1,1)\right)=\rho/2$, so it becomes

$$\frac{4}{\rho^2K^2}\,\mathrm{Var}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}-Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\})\right). \tag{43}$$

For ease of exposition, we always hide the constant until the last step. We first process the terms with only $Y_t(\boldsymbol{W}_{1:t-2},1,1)$ by conditioning on $\boldsymbol{W}_{t-1:t}$. Based on the law of total variance, we have

$$\begin{aligned}
&\mathrm{Var}\left(\sum_{t\in\mathcal{T}_b} Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}\right)\\
=&\mathrm{Var}\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_b} Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right]\right)\\
&+\mathbb{E}\left[\mathrm{Var}\left(\sum_{t\in\mathcal{T}_b} Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right)\right].
\end{aligned} \tag{44}$$

This is exactly the sum of (i-a) and (ii-a) in the sketch. Then, the first part (i-a) is equivalent to

$$\begin{aligned}
&\mathrm{Var}\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_b} Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right]\right)\\
=&\sum_{t\in\mathcal{T}_b}(\mathbb{P}(\boldsymbol{W}_{t-1:t}=(1,1))-\mathbb{P}(\boldsymbol{W}_{t-1:t}=(1,1))^2)\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]^2\\
&+\sum_{t\neq t'\in\mathcal{T}_b}(\mathbb{P}(\boldsymbol{W}_{t-1:t}=(1,1),\boldsymbol{W}_{t'-1:t'}=(1,1))-\mathbb{P}(\boldsymbol{W}_{t-1:t}=(1,1))\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1)))\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)]\\
=&\sum_{t\in\mathcal{T}_b}(\frac{\rho}{2}-\frac{\rho^2}{4})\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]^2+\sum_{t\neq t'\in\mathcal{T}_b}(\frac{\rho(K_v/2-1)}{2(K-1)}-\frac{\rho^2}{4})\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)]\\
=&\sum_{t\in\mathcal{T}_b}\frac{(2-\rho)\rho}{4}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]^2-\sum_{t\neq t'\in\mathcal{T}_b}\frac{(2-\rho)\rho}{4(K-1)}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)]\\
=&\sum_{t\in\mathcal{T}_b}\left(\frac{(2-\rho)\rho}{4}+\frac{(2-\rho)\rho}{4(K-1)}\right)\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]^2-\sum_{t,t'\in\mathcal{T}_b}\frac{(2-\rho)\rho}{4(K-1)}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)]\\
=&\frac{(2-\rho)\rho K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})^2.
\end{aligned} \tag{45}$$

where the first equality decomposes the sum into the cases $t=t'$ and $t\neq t'$; the second equality computes the probabilities based on our random permutation at each block ($\rho=K_v/K$ and $K_v/2$ is the number of pairs $(1,1)$); the last equality aggregates the second part into the average outcomes within the block.[7]

A symmetric result can be obtained for the part (i-b) including $Y_t(\boldsymbol{W}_{1:t-2},0,0)$. Then we look at the interaction between $Y_t(\boldsymbol{W}_{1:t-2},1,1)$ and $Y_t(\boldsymbol{W}_{1:t-2},0,0)$. From the law of total covariance, we have

[7] Throughout the proof, when we present $\mathbb{E}[[Y_t(\boldsymbol{W}_{1:t-2},1,1)]]$, the expectation is always taken over the conditional distribution of $\boldsymbol{W}_{1:t-2}$ given $\boldsymbol{W}_{t-1:t}=(1,1)$. Similarly, when it comes to $\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))$, the covariance is always taken over the conditional distribution given $\boldsymbol{W}_{t-1:t}=(1,1),\boldsymbol{W}_{t'-1:t'}=(1,1)$.

$$
\begin{aligned}
&-2\operatorname{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\},\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}\right)\\
=&-2\operatorname{Cov}\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right],\mathbb{E}\left[\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right]\right)\\
&-2\mathbb{E}\left[\operatorname{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\},\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}\right)|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right].
\end{aligned}
\tag{46}
$$

This is exactly the sum of (i-c) and (ii-c) in the sketch. Then the first part (i-c) is equivalent to

$$
\begin{aligned}
&-2\operatorname{Cov}\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right],\mathbb{E}\left[\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right]\right)\\
=&\sum_{t\in\mathcal{T}_b}\frac{\rho^2}{2}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]+\sum_{t\neq t'\in\mathcal{T}_b}(\frac{\rho^2}{2}-\frac{\rho^2K}{2(K-1)})\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)]\\
=&\sum_{t\in\mathcal{T}_b}\frac{\rho^2}{2}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\sum_{t\neq t'\in\mathcal{T}_b}\frac{\rho^2}{2(K-1)}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)].\\
=&\frac{\rho^2K}{2(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]}).
\end{aligned}
\tag{47}
$$

Now we have derived the three pieces related to the variance of conditional expectation: (i-a), (i-b) and (i-c). It remains to analyze the expectation of conditional variance. The part (ii-a) is equivalent to

$$
\begin{aligned}
&\mathbb{E}\left[\operatorname{Var}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right)\right]\\
=&\sum_{t\in\mathcal{T}_b}\frac{\rho}{2}\operatorname{Var}(Y_t(\boldsymbol{W}_{1:t-2},1,1))+\sum_{t\neq t'\in\mathcal{T}_b}\frac{\rho(K_v/2-1)}{2(K-1)}\operatorname{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)).
\end{aligned}
\tag{48}
$$

A symmetric result can be obtained for the part (ii-b) including $Y_t(\boldsymbol{W}_{1:t-2},0,0)$. Moreover, the part (ii-c) including the interaction between $Y_t(\boldsymbol{W}_{1:t-2},1,1)$ and $Y_t(\boldsymbol{W}_{1:t-2},0,0)$ is equivalent to

$$
\begin{aligned}
&-2\mathbb{E}\left[\operatorname{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\},\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}\right)|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right]\\
=&-2\sum_{t\neq t'\in\mathcal{T}_b}\frac{\rho^2K}{4(K-1)}\operatorname{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)).
\end{aligned}
\tag{49}
$$

Finally, the variance within a block is given by

$$
\begin{aligned}
&\mathrm{Var}(\widehat{\tau}_b)\\
=&\frac{(2-\rho)}{\rho K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})^2+(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})^2\\
&+\frac{2}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})\\
&+\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-2},1,1))+\frac{K_v-2}{\rho K(K-1)}\sum_{t\neq t'\in\mathcal{T}_b}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))\\
&-\sum_{t\neq t'\in\mathcal{T}_b}\frac{2}{K(K-1)}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0))\\
=&\frac{2}{\rho K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})^2+(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})^2\\
&-\frac{1}{K(K-1)}\sum_{t\in\mathcal{T}_b}\Big((\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)])-(\overline{\mathbb{E}[Y_b(\cdot,1,1)]}-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})\Big)^2\\
&+\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-2},1,1))+\frac{K_v-2}{\rho K^2(K-1)}\sum_{t\neq t'\in\mathcal{T}_b}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))\\
&+\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-2},0,0))+\frac{K_v-2}{\rho K^2(K-1)}\sum_{t\neq t'\in\mathcal{T}_b}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0))\\
&-\sum_{t\neq t'\in\mathcal{T}_b}\frac{2}{K(K-1)}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)).
\end{aligned}
\tag{50}
$$

where the first equality transforms the interaction terms into the variance of treatment outcomes, the variance of control outcomes, and the variance of effect.

The Case $|b'-b|=1$.

Next, we study the covariance between two distinct blocks. Let us start with $\delta_{b,b'}=|b-b'|=1$ and highlight the part that differs from the previous case.

$$
\begin{aligned}
\mathrm{Cov}(\widehat{\tau}_b,\widehat{\tau}_{b'})=\frac{4}{\rho^2K^2}\mathrm{Cov}\Bigg(&\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\{\boldsymbol{W}_{t-1:t}=(1,1)\}-Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\{\boldsymbol{W}_{t-1:t}=(0,0)\}),\\
&\sum_{t'\in\mathcal{T}_{b'}}(Y_t(\boldsymbol{W}_{1:t'-2},1,1)\mathbb{1}\{\boldsymbol{W}_{t'-1:t'}=(1,1)\}-Y_t(\boldsymbol{W}_{1:t'-2},0,0)\mathbb{1}\{\boldsymbol{W}_{t'-1:t'}=(0,0)\})\Bigg).
\end{aligned}
\tag{51}
$$

Similar to the analysis above, we will decompose it into six components: (i-a), (i-b), (i-c), (ii-a), (ii-b) and (ii-c). For the covariance of treatment outcomes, we apply the law of total covariance:

$$
\begin{aligned}
&\operatorname{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\},\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(1,1)\right\}\right)\\
=&\operatorname{Cov}\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t}\right],\mathbb{E}\left[\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(1,1)\right\}|\boldsymbol{W}_{t'-1:t'}\right]\right)\\
&+\mathbb{E}\left[\operatorname{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\},\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(1,1)\right\}|\boldsymbol{W}_{t'-1:t'},\boldsymbol{W}_{t-1:t}\right)\right].
\end{aligned}
\tag{52}
$$

The first part (i-a) is equivalent to

$$
\begin{aligned}
&\sum_{t\in\mathcal{T}_b}\left(\frac{\rho^2}{2}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-1},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]-\frac{\rho^2}{4}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-1},1,1)]\right)\\
+&\sum_{\substack{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b+1}\\t'\neq t+1}}\left(\frac{\rho^2(K/2-1)}{2(K-1)}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]-\frac{\rho^2}{4}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)]\right)\\
=&\sum_{t\in\mathcal{T}_b}\frac{\rho^2}{2}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-1},1,1)|\boldsymbol{W}_{t-1:t}=(1,1)]\\
&+\sum_{\substack{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b+1}\\t'\neq t+1}}\frac{\rho^2(K/2-1)}{2(K-1)}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]\\
&-\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b+1}}\frac{\rho^2}{4}\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)].
\end{aligned}
\tag{53}
$$

Note that compared to the case of $b=b'$ in (45), the key difference here is that we need to deal with the conditional expectation $\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]$ because the assignments are dependent across blocks. To see this, when $t'=t+1$, we easily have $\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]=\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},1,1,1)]$; when $t'\neq t+1$ (but t' and $t+1$ are still from the same block), according to our design, we have

$$
\begin{aligned}
&\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]\\
=&\mathbb{P}(\boldsymbol{W}_{t'-2}=1|\boldsymbol{W}_{t-1:t}=(1,1))\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},1,1,1)]+\mathbb{P}(\boldsymbol{W}_{t'-2}=0|\boldsymbol{W}_{t-1:t}=(1,1))\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},0,1,1)]\\
=&\frac{\rho K/2-1}{K/2-1}\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},1,1,1)]+\frac{(1-\rho)K/2}{K/2-1}\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},0,1,1)]
\end{aligned}
\tag{54}
$$

In this way, (53) can be reformulated into

$$
\begin{aligned}
&\left(\frac{\rho^2}{2}-\frac{\rho^3}{4}\right)\frac{K}{K-1}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},1,1,1)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,1,1,1)]})\\
-&\frac{\rho^2(1-\rho)}{4}\frac{K}{K-1}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},0,1,1)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,1,1)]}).
\end{aligned}
\tag{55}
$$

A symmetric derivation applies to the part (i-b) including $Y_t(\boldsymbol{W}_{1:t-2},0,0)$. For the part (i-c) including the interactions, we have

$$
\begin{aligned}
&- \operatorname{Cov}\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right],\mathbb{E}\left[\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(0,0)\right\}|\boldsymbol{W}_{t'-1:t'},\forall t'\in\mathcal{T}_{b'}\right]\right)\\
&- \operatorname{Cov}\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}|\boldsymbol{W}_{t-1:t},\forall t\in\mathcal{T}_b\right],\mathbb{E}\left[\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(1,1)\right\}|\boldsymbol{W}_{t'-1:t'},\forall t'\in\mathcal{T}_{b'}\right]\right)\\
=&\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b+1}}\frac{\rho^2}{4}\left(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)]+\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)]\right)\\
-&\sum_{\substack{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b+1}\\t'\neq t+1}}\frac{\rho^2K}{4(K-1)}\left(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0|\boldsymbol{W}_{t-1:t}=(1,1))]+\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(0,0))]\right)\\
=&\frac{\rho^3K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},0,0,0)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,0,0)]})\\
+&\frac{\rho^2(1-\rho)K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},1,0,0)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,1,0,0)]})\\
+&\frac{\rho^3K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},1,1,1)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,1,1,1)]})\\
+&\frac{\rho^2(1-\rho)K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},0,1,1)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,1,1)]})\\
=&\frac{\rho^3K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},1,1,1)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,1,1,1)]})\\
+&\frac{\rho^3K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},0,0,0)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,0,0)]})\\
+&\frac{\rho^2(1-\rho)K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},0,1,1)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,1,1)]})\\
+&\frac{\rho^2(1-\rho)K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},1,0,0)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,1,0,0)]})\\
-&\frac{\rho^2K}{4(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-(\overline{\mathbb{E}[Y_b(\cdot,1,1)]}-\overline{\mathbb{E}[Y_b(\cdot,0,0)]}))\\
&\cdot(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-1},1,1)]-\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-1},0,0)]-(\overline{\mathbb{E}[Y_{b+1}(\cdot,1,1)]}-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,0)]})).
\end{aligned}
\tag{56}
$$

Then it remains to analyze the expectation of covariance. First the part (ii-a) including $Y_t(\boldsymbol{W}_{1:t-2},1,1)$ is equivalent to

$$
\begin{aligned}
&\mathbb{E}\left[\operatorname{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(1,1)\right\}|\boldsymbol{W}_{t'-1:t'}\right)\right]\\
=&\sum_{t\in\mathcal{T}_b}\frac{\rho^2}{2}\operatorname{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t+1}(\boldsymbol{W}_{1:t-2},1,1,1)\right)\\
&+\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'},t'\neq t+1}\frac{\rho^2(K-2)}{4(K-1)}\operatorname{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\right).
\end{aligned}
\tag{57}
$$

A symmetric result can be obtained for the part (ii-b) including $Y_t(\boldsymbol{W}_{1:t-2},0,0)$. Moreover, the part (ii-c) including the interactions is equivalent to

$$
\begin{aligned}
&-\mathbb{E}\left[\mathrm{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(1,1)\right\}|\boldsymbol{W}_{t-1:t},\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(0,0)\right\}|\boldsymbol{W}_{t'-1:t'}\right)\right]\\
&-\mathbb{E}\left[\mathrm{Cov}\left(\sum_{t\in\mathcal{T}_b}(Y_t(\boldsymbol{W}_{1:t-2},0,0)\mathbb{1}\left\{\boldsymbol{W}_{t-1:t}=(0,0)\right\}|\boldsymbol{W}_{t-1:t},\sum_{t'\in\mathcal{T}_{b'}}(Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\mathbb{1}\left\{\boldsymbol{W}_{t'-1:t'}=(1,1)\right\}|\boldsymbol{W}_{t'-1:t'}\right)\right]\\
=&-\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'},t'\neq t+1}\frac{\rho^2K}{4(K-1)}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0))\\
&-\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'},t'\neq t+1}\frac{\rho^2K}{4(K-1)}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)).
\end{aligned}
\tag{58}
$$

Combining all six components together, the covariance between two adjacent blocks becomes

$$
\begin{aligned}
&\mathrm{Cov}(\widehat{\tau}_b,\widehat{\tau}_{b+1})\\
=&\frac{2}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},1,1,1)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,1,1,1)]})\\
+&\frac{2}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-2},0,0,0)]-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,0,0)]})\\
-&\frac{1}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-(\overline{\mathbb{E}[Y_b(\cdot,1,1)]}-\overline{\mathbb{E}[Y_b(\cdot,0,0)]}))\\
&\cdot(\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-1},1,1)]-\mathbb{E}[Y_{t+1}(\boldsymbol{W}_{1:t-1},0,0)]-(\overline{\mathbb{E}[Y_{b+1}(\cdot,1,1)]}-\overline{\mathbb{E}[Y_{b+1}(\cdot,0,0)]}))\\
+&\frac{2}{K^2}\sum_{t\in\mathcal{T}_b}(\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t+1}(\boldsymbol{W}_{1:t-2},1,1,1)\right)+\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t+1}(\boldsymbol{W}_{1:t-2},0,0,0)\right))\\
+&\frac{\rho^2(K-2)}{4(K-1)}\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'},t'\neq t+1}(\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\right)+\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\right))\\
-&\frac{\rho^2K}{4(K-1)}\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'},t'\neq t+1}(\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\right)+\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\right)).
\end{aligned}
\tag{59}
$$

General Cases.

To extend the analysis to the case when $\delta_{b,b'}=|b-b'|>1$, we can repeat the same way of decomposition and characterize the coefficients using

$$
\begin{aligned}
\gamma^+_{t,t'}&=\frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1)|\boldsymbol{W}_{t-1:t}=(1,1))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1))}=\frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0)|\boldsymbol{W}_{t-1:t}=(0,0))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0))},\\
\gamma^-_{t,t'}&=\frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0)|\boldsymbol{W}_{t-1:t}=(1,1))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(0,0))}=\frac{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1)|\boldsymbol{W}_{t-1:t}=(0,0))}{\mathbb{P}(\boldsymbol{W}_{t'-1:t'}=(1,1))}.
\end{aligned}
\tag{60}
$$

Specifically, the covariance of conditional expectation (i.e., the sum of (i-a), (i-b), and (i-c)) is given by

$$
\begin{aligned}
&\frac{\gamma^{+}_{t,t+\delta_{b,b'}}}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]-\overline{\mathbb{E}[Y_{b'}(\cdot,1,1|\boldsymbol{W}_{t-1:t}=(1,1))]})\\
+&\frac{\gamma^{+}_{t,t+\delta_{b,b'}}}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},0,0|\boldsymbol{W}_{t-1:t}=(0,0))]-\overline{\mathbb{E}[Y_{b'}(\cdot,0,0|\boldsymbol{W}_{t-1:t}=(0,0))]})\\
-&\frac{\gamma^{-}_{t,t+\delta_{b,b'}}}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},0,0|\boldsymbol{W}_{t-1:t}=(1,1))]-\overline{\mathbb{E}[Y_{b'}(\cdot,0,0|\boldsymbol{W}_{t-1:t}=(1,1))]})\\
-&\frac{\gamma^{-}_{t,t+\delta_{b,b'}}}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},1,1|\boldsymbol{W}_{t-1:t}=(0,0))]-\overline{\mathbb{E}[Y_{b'}(\cdot,1,1|\boldsymbol{W}_{t-1:t}=(0,0))]})\\
-&\frac{1}{K(K-1)}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-(\overline{\mathbb{E}[Y_b(\cdot,1,1)]}-\overline{\mathbb{E}[Y_b(\cdot,0,0)]}))\\
&\cdot(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},1,1)]-\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},0,0)]-(\overline{\mathbb{E}[Y_{b'}(\cdot,1,1)]}-\overline{\mathbb{E}[Y_{b'}(\cdot,0,0)]})).
\end{aligned}
\tag{61}
$$

By defining $\bar{\tau}_k=\frac{1}{B}\sum_{b=1}^{B}(\mathbb{E}[Y_{(k-1)B+b}(\boldsymbol{W}_{1:(k-1)B+b-2},1,1)]-\mathbb{E}[Y_{(k-1)B+b}(\boldsymbol{W}_{1:(k-1)B+b-2},0,0)])$ to be the average treatment effect in the k-th day, and for $u\in\{0,1\},v\in\{0,1\}$,

$$
\begin{aligned}
S^{u,v}_{b,b'}=\frac{1}{K-1}\sum_{t\in\mathcal{T}_b}&(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},\boldsymbol{u}_2)]-\overline{\mathbb{E}[Y_b(\cdot,\boldsymbol{u}_2)]})\\
&\cdot(\mathbb{E}[Y_{t+\delta_{b,b'}}(\boldsymbol{W}_{1:t+\delta_{b,b'}-2},\boldsymbol{v}_2|\boldsymbol{W}_{t-1:t}=\boldsymbol{u}_2)]-\overline{\mathbb{E}[Y_{b'}(\cdot,\boldsymbol{v}_2|\boldsymbol{W}_{t-1:t}=\boldsymbol{u}_2)]}).
\end{aligned}
\tag{62}
$$

we aggregate all block-wise covariance of conditional expectation and obtain

$$
V_1=\frac{1}{B^2K}\sum_{b,b'\in[B]}(\gamma^{+}_{b,b'}(S^{1,1}_{b,b'}+S^{0,0}_{b,b'})-\gamma^{-}_{b,b'}(S^{1,0}_{b,b'}+S^{0,1}_{b,b'}))-\frac{1}{K(K-1)}\sum_{k=1}^{K}(\bar{\tau}_k-\bar{\tau})^2. \tag{63}
$$

Next, the expectation of conditional covariance (i.e., the sum of (ii-a), (ii-b), and (ii-c)) is given by

$$
\begin{aligned}
&\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}\frac{\gamma^{+}_{t,t'}}{K^2}(\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\right)+\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\right))\\
-&\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}\frac{\gamma^{-}_{t,t'}}{K^2}(\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\right)+\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\right))
\end{aligned}
\tag{64}
$$

By defining for $u\in\{0,1\},v\in\{0,1\}$,

$$
\sigma^{u,v}_{t,t'}=\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},\boldsymbol{u}_2),Y_{t'}(\boldsymbol{W}_{1:t'-2},\boldsymbol{v}_2)\right), \tag{65}
$$

we aggregate all block-wise expectation of conditional covariance and obtain

$$
V_2=\frac{1}{B^2K^2}\sum_{b,b'\in[B]}\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}(\gamma^{+}_{t,t'}(\sigma^{1,1}_{t,t'}+\sigma^{0,0}_{t,t'})-\gamma^{-}_{t,t'}(\sigma^{1,0}_{t,t'}+\sigma^{0,1}_{t,t'})). \tag{66}
$$

Recursion of Coefficients.

Lastly, we show that for $t \in \mathcal{T}_b$ and $t' \in \mathcal{T}_{b'}$, the coefficients $\gamma^+_{t,t'}$ and $\gamma^-_{t,t'}$ can be derived by recursion. Based on our design, we see that they depend on two factors: the distance $\delta_{b,b'}$ between two blocks and whether two periods belong to the same day (i.e., $|t'-t| = \delta_{b,b'}$). We take the case when $|t'-t| = \delta_{b,b'}$ as an instance to illustrate how recursion works as we increase $\delta_{b,b'}$, and the other case follows the same procedure.

For ease of notation, we define

$$q_{uv,u'v'}(\delta) = \mathbb{P}(\boldsymbol{W}_{t-1:t} = (u,v) \cap \boldsymbol{W}_{t'-1:t'} = (u,v)) \tag{67}$$

Let us fix t and let $t' = t + \delta$. As we increase δ by 1, we have

$$\begin{aligned}
q_{11,11}(\delta) =& (q_{11,11}(\delta-1) + q_{11,01}(\delta-1))\rho \\
q_{11,00}(\delta) =& (q_{11,00}(\delta-1) + q_{11,10}(\delta-1))\rho \\
q_{11,01}(\delta) =& (q_{11,00}(\delta-1) + q_{11,10}(\delta-1))(1-\rho) \\
q_{11,10}(\delta) =& (q_{11,11}(\delta-1) + q_{11,01}(\delta-1))(1-\rho)
\end{aligned} \tag{68}$$

with

$$q_{11,11}(1) = \frac{\rho^2}{2}, q_{11,00}(1) = 0, q_{11,01}(1) = 0, q_{11,10}(1) = \frac{\rho(1-\rho)}{2}. \tag{69}$$

Finally, the overlapping indices between blocks are given by

$$\gamma^+_{b,b'} = \gamma^+_{t,t+\delta_{b,b'}} = \frac{4q_{11,11}(\delta_{b,b'})}{\rho^2}, \gamma^-_{b,b'} = \gamma^-_{t,t+\delta_{b,b'}} = \frac{4q_{11,00}(\delta_{b,b'})}{\rho^2}. \tag{70}$$

D. Proof of Corollary 1

Similar to our analysis in variance, we start with the case $b = b'$ to show the idea. Based on our derivation in (50), the expectation of the conditional covariance is

$$\begin{aligned}
&\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-2},1,1)) + \frac{\rho K-2}{\rho K^2(K-1)}\sum_{t\neq t'\in\mathcal{T}_b}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1), Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)) \\
&+\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-2},0,0)) + \frac{\rho K-2}{\rho K^2(K-1)}\sum_{t\neq t'\in\mathcal{T}_b}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},0,0), Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)) \\
&-\frac{2}{K(K-1)}\sum_{t\neq t'\in\mathcal{T}_b}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1), Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)).
\end{aligned} \tag{71}$$

We will analyze the first row by giving an upper bound with Assumption 2, and the analysis of the second and the third row follows similarly. The key idea is to sequentially apply the law of total variance to $W_{t-2}, W_{t-3}, \ldots$.

First, by conditioning on the assignments W_{t-2}, we have

$$\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-2},1,1)) = \mathrm{Var}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)|W_{t-2}]) + \rho\,\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-3},1,1,1)) + (1-\rho)\,\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-3},0,1,1)). \tag{72}$$

Similarly, by conditioning on the assignments W_{t-2} and $W_{t'-2}$, we have

$$\begin{aligned}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)) =& \mathrm{Cov}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)|W_{t-2}],\mathbb{E}[Y_t(\boldsymbol{W}_{1:t'-2},1,1)|W_{t'-2}]) \\ &+\mathbb{E}[\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))|W_{t-2},W_{t'-2}].\end{aligned} \tag{73}$$

Based on our design, we have

$$\mathrm{Var}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)|W_{t-2}]) = \rho(1-\rho)(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)] - \mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1))])^2. \tag{74}$$

Similarly, we have

$$\begin{aligned}&\mathrm{Cov}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)|W_{t-2}],\mathbb{E}[Y_t(\boldsymbol{W}_{1:t'-2},1,1)|W_{t'-2}]) \\ =&-\frac{\rho(1-\rho)}{K/2-1}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)] - \mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1))])(\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},1,1,1)] - \mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},0,1,1))]).\end{aligned} \tag{75}$$

The coefficient is negative because conditional on $W_{t-1} = W_{t'-1} = 1$, the assignment probabilities of W_{t-2} and $W_{t'-2}$ are negatively correlated. Then we first aggregate (74) and (75) into the first row of (71):

$$\begin{aligned}&\frac{2(1-\rho)}{K^2}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)] - \mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1)])^2 \\ &-\frac{(\rho K-2)(1-\rho)}{K^2(K-1)(K/2-1)}\sum_{t\neq t'\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)] - \mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1)])(\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},1,1,1)] - \mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-3},0,1,1)]) \\ \le&\left(\frac{2(1-\rho)}{K^2}+\frac{(\rho K-2)(1-\rho)}{K^2(K-1)(K/2-1)}\right)\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)] - \mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1)])^2 \\ \le&\frac{2(1-\rho)}{K^2}\left(1+\frac{\rho K/2-1}{(K-1)(K/2-1)}\right)\sum_{t\in\mathcal{T}_b}\phi_t(2)^2 \le \frac{2(1-\rho)}{K^2}\left(1+\frac{\rho}{K-1}\right)\sum_{t\in\mathcal{T}_b}\phi_t(2)^2\end{aligned} \tag{76}$$

Using the same conditioning on W_{t-3}, we can further analyze the remaining parts in (72) and (73). Taking the terms including $Y_t(\boldsymbol{W}_{t-3},1,1,1)$ as instance, in (72), we have

$$\begin{aligned}&\rho\,\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-3},1,1,1)) \\ =&\rho\left(\mathrm{Var}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)|W_{t-3}]) + \rho\,\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-4},1,1,1,1)) + (1-\rho)\,\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-4},0,1,1,1))\right)\end{aligned} \tag{77}$$

and in (73), we have

$$\begin{aligned}&\frac{\rho(\rho K/2-1)}{K/2-1}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-3},1,1,1),Y_{t'}(\boldsymbol{W}_{1:t'-3},1,1,1)) \\ =&\frac{\rho(\rho K/2-1)}{K/2-1}\mathrm{Cov}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)|W_{t-3}],\mathbb{E}[Y_t(\boldsymbol{W}_{1:t'-3},1,1,1)|W_{t'-3}]) \\ &+\frac{\rho(\rho K/2-1)}{K/2-1}\mathbb{E}[\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-3},1,1,1),Y_{t'}(\boldsymbol{W}_{1:t'-3},1,1,1))|W_{t-3},W_{t'-3}].\end{aligned} \tag{78}$$

For both equations, we take out their first parts, follow the same derivation in (74) and (75), and aggregate them together. This gives

$$
\begin{aligned}
&\frac{2(1-\rho)\rho}{K^2}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-4},\mathbf{1}_4)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-4},0,1,1,1))])^2\\
&-\frac{\rho(1-\rho)(\rho K-2)^2}{2K^2(K-1)(K/2-1)^2}\sum_{t\neq t'\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-4},\mathbf{1}_4)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-4},0,1,1,1)])(\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-4},\mathbf{1}_4)]-\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-4},0,1,1,1)])\\
&\leq\left(\frac{2(1-\rho)\rho}{K^2}+\frac{\rho(1-\rho)(\rho K-2)^2}{2K^2(K-1)(K/2-1)^2}\right)\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-4},\mathbf{1}_4)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-4},0,1,1,1)])^2\\
&\leq\frac{2(1-\rho)\rho}{K^2}\left(1+\frac{(\rho K/2-1)^2}{(K-1)(K/2-1)^2}\right)\sum_{t\in\mathcal{T}_b}\phi_t(3)^2\leq\frac{2(1-\rho)\rho}{K^2}\left(1+\frac{\rho^2}{K-1}\right)\sum_{t\in\mathcal{T}_b}\phi_t(3)^2.
\end{aligned}
\tag{79}
$$

For the terms including $Y_t(\boldsymbol{W}_{t-3},0,1,1)$, we have another upper bound

$$
\frac{2(1-\rho)^2}{K^2}\left(1+\frac{\rho(1-\rho)}{K-1}\right)\sum_{t\in\mathcal{T}_b}\phi_t(3)^2. \tag{80}
$$

Thus, the sum of them is bounded by

$$
\frac{2(1-\rho)}{K^2}\left(1+\frac{\rho(1-\rho)^2+\rho^3}{K-1}\right)\sum_{t\in\mathcal{T}_b}\phi_t(3)^2\leq\frac{2(1-\rho)}{K^2}\left(1+\frac{\rho}{K-1}\right)\sum_{t\in\mathcal{T}_b}\phi_t(3)^2. \tag{81}
$$

By repeating this sequential conditioning, we finally bound the first row of (71) by

$$
\frac{2(1-\rho)}{K^2}\left(1+\frac{\rho}{K-1}\right)\sum_{t\in\mathcal{T}_b}\sum_{q=2}^{\infty}\phi_t(q)^2\leq\frac{2(1-\rho)}{K}\left(1+\frac{\rho}{K-1}\right)\sum_{q=2}^{\infty}\phi(q)^2\leq\frac{2(1-\rho)}{K-1}\sum_{q=2}^{\infty}\phi(q)^2, \tag{82}
$$

A similar bound can be obtained for the second row of (71). Lastly, following the same procedure, we bound the third row of (71):

$$
\begin{aligned}
&-\frac{2}{K(K-1)}\sum_{t\neq t'\in\mathcal{T}_b}\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0))\\
\leq&\frac{2}{K(K-1)}\frac{\rho(1-\rho)}{K/2-1}\sum_{t\neq t'\in\mathcal{T}_b}\sum_{q=3}^{\infty}\phi_t(q)\phi_{t'}(q)\leq\frac{2\rho(1-\rho)}{K/2-1}\sum_{q=3}^{\infty}\phi(q)^2.
\end{aligned}
\tag{83}
$$

Altogether, (71) has an upper bound

$$
\frac{4(1-\rho)}{K-1}\sum_{q=2}^{\infty}\phi(q)^2+\frac{2\rho(1-\rho)}{K/2-1}\sum_{q=3}^{\infty}\phi(q)^2\leq\frac{4(1-\rho)}{K-1}\phi(2)^2+\frac{4(1-\rho^2)}{K-2}\sum_{q=3}^{\infty}\phi(q)^2 \tag{84}
$$

Finally, for the general case with $\delta_{b,b'}\geq 1$, we know from (64) that the expectation of conditional covariance can be represented as

$$
\begin{aligned}
&\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}\frac{\gamma^+_{t,t'}}{K^2}(\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\right)+\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\right))\\
&-\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}\frac{\gamma^-_{t,t'}}{K^2}(\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0)\right)+\mathrm{Cov}\left(Y_t(\boldsymbol{W}_{1:t-2},0,0),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)\right))
\end{aligned}
\tag{85}
$$

Following the same sequential conditioning process to decompose each covariance term, we get an upper bound

$$\begin{aligned}
&\frac{2(\gamma^+_{b,b'}+\gamma^-_{b,b'})\rho(1-\rho)}{K^2}\sum_{q=2}^{\infty}\sum_{t\in\mathcal{T}_b}\phi_t(q)\phi_{t+\delta_{b,b'}}(q+\delta_{b,b'})\\
&\quad+\frac{4\rho(1-\rho)}{K^2(K-1)}\sum_{t\in\mathcal{T}_b}\sum_{t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\phi_t(2)\phi_{t'}(2+\delta_{b,b'})\\
+&\frac{2(\gamma^+_{t,t'}+\gamma^-_{t,t'})\rho(1-\rho)}{K^2(K/2-1)}\sum_{q=3}^{\infty}\sum_{t\in\mathcal{T}_b}\sum_{t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\phi_t(q)\phi_{t'}(q+\delta_{b,b'}).
\end{aligned}\tag{86}$$

Based on our recursion in Appendix C, for $\delta_{b,b'}\geq 1$, we actually have a closed-form expression $\gamma^+_{b,b'}+\gamma^-_{b,b'}=2$ and $\gamma^+_{t,t'}+\gamma^-_{t,t'}=2$, which gives rise to

$$\begin{aligned}
&\frac{4\rho(1-\rho)}{K^2}\sum_{q=2}^{\infty}\sum_{t\in\mathcal{T}_b}\phi_t(q)\phi_{t+\delta_{b,b'}}(q+\delta_{b,b'})+\frac{4\rho(1-\rho)}{K^2(K-1)}\sum_{t\in\mathcal{T}_b}\sum_{t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\phi_t(2)\phi_{t'}(2+\delta_{b,b'})\\
&+\frac{4\rho(1-\rho)}{K^2(K/2-1)}\sum_{q=3}^{\infty}\sum_{t\in\mathcal{T}_b}\sum_{t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\phi_t(q)\phi_{t'}(q+\delta_{b,b'})\\
\leq&\frac{4\rho(1-\rho)}{K}\sum_{q=2}^{\infty}\phi(q)\phi(q+\delta_{b,b'})+\frac{4\rho(1-\rho)}{K}\phi(2)\phi(2+\delta_{b,b'})+\frac{4(K-1)\rho(1-\rho)}{K(K/2-1)}\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'})\\
=&\frac{8\rho(1-\rho)}{K}\phi(2)\phi(2+\delta_{b,b'})+\frac{4\rho(1-\rho)(3K/2-2)}{K(K/2-1)}\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'})
\end{aligned}\tag{87}$$

E. Proof of Corollary 2

We first elaborate on Assumption 3. We define several short-hand notations:

$$\begin{aligned}
\tau_t(11-00)=&\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)],\\
\tau_{t'|t}(11-00|11)=&\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(1,1))]-\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0|\boldsymbol{W}_{t-1:t}=(1,1))],\\
\tau_{t'|t}(11-00|00)=&\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1|\boldsymbol{W}_{t-1:t}=(0,0))]-\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0|\boldsymbol{W}_{t-1:t}=(0,0))].
\end{aligned}\tag{88}$$

and

$$\begin{aligned}
\Delta^1_t=&(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)]-\overline{\mathbb{E}[Y_b(\cdot,1,1)]})(\tau_{t+\delta_{b,b'}}(11-00)-\tau_{t+\delta_{b,b'}|t}(11-00|11)-(\overline{\tau_{b'}(11-00)}-\overline{\tau_{b'|b}(11-00|11)}),\\
\Delta^0_t=&(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},0,0)]-\overline{\mathbb{E}[Y_b(\cdot,0,0)]})(\tau_{t+\delta_{b,b'}}(11-00)-\tau_{t+\delta_{b,b'}|t}(11-00|00)-(\overline{\tau_{b'}(11-00)}-\overline{\tau_{b'|b}(11-00|00)})
\end{aligned}\tag{89}$$

By expanding the covariance, the equality in the assumption is equivalent to

$$\sum_{t\in\mathcal{T}_b}(\Delta^1_t-\Delta^0_t)=0.\tag{90}$$

There are several sufficient conditions. First, if the carryover effect is limited to one period (Bojinov et al. 2022, Chen and Simchi-Levi 2023), then $\tau_{t+\delta_{b,b'}}(11-00)\equiv\tau_{t+\delta_{b,b'}|t}(11-00|00)\equiv\tau_{t+\delta_{b,b'}|t}(11-$

$00|11)$ so $\Delta_t^1 = \Delta_t^0 = 0$. Second, if the treatment effect is constant across periods in the same block, then $\tau_{t+\delta_{b,b'}}(11-00) \equiv \overline{\tau_{b'}(11-00)}$ and $\tau_{t+\delta_{b,b'}|t}(11-00|11) \equiv \overline{\tau_{b'|b}(11-00|11)}$ so we still have $\Delta_t^1 = \Delta_t^0 = 0$. Third, if past treatment assignments do not impact the future treatment effect, then $\tau_{t+\delta_{b,b'}}(11-00) \equiv \tau_{t+\delta_{b,b'}}(11-00|11)$. That is, the treatment assignments $\boldsymbol{W}_{t-1:t} = (1,1)$ may shift the outcomes we could observe at period $t+\delta_{b,b'}$ under $\boldsymbol{W}_{t+\delta_{b,b'}-1:t+\delta_{b,b'}} = (1,1)$ and $\boldsymbol{W}_{t+\delta_{b,b'}-1:t+\delta_{b,b'}} = (0,0)$ (i.e., the first condition fails), but the difference remains the same. This is true under many outcome models, such as the time series in (3) and any generalized linear models, but may not be true under certain cases such as the presence of the habituation/cumulative effect (Basse et al. 2023), referring to the phenomenon where, through repeated or prolonged exposure to a particular treatment, the response gradually diminishes. Fourth, in the absence of the previous conditions, it is likely that $\tau_{t+\delta_{b,b'}}(11-00) - \tau_{t+\delta_{b,b'}|t}(11-00|11) \neq 0$ due to the habituation effect. Nonetheless, we can still have $\Delta_t^1 = \Delta_t^0 = 0$ as long as the habituation effect does not differentiate by day (i.e. $\tau_{t+\delta_{b,b'}}(11-00) - \tau_{t+\delta_{b,b'}|t}(11-00|11) \equiv \overline{\tau_{b'}(11-00)} - \overline{\tau_{b'|b}(11-00|11)}$). Lastly, we believe that the assumption may hardly be perfectly satisfied in reality because it is almost impossible to characterize the real-world dynamics using well-studied models that meet the sufficient conditions. Nonetheless, the violation of the equality should be mostly negligible in practice. Kohavi et al. (2020) point out that in business experimentation, most experiments yield small effects because many interventions are typically small product/algorithm changes. Therefore, the scale of the treatment effect variation is often way smaller than the treatment/control outcome variation, and scale of the habituation effect is even smaller. In other words, $|\sum_{t\in\mathcal{T}_b}(\Delta_t^1 - \Delta_t^0)| << V_1$.

Under Assumption 3, we have that

$$S_{b,b'}^{1,0} + S_{b,b'}^{0,1} = S_{b,b'}^{1,1} + S_{b,b'}^{0,0} - S_{b,b'}^{\tau}. \tag{91}$$

Thus, for each pair of two blocks b and b', the first part of the variance is equivalent to

$$\begin{aligned} &\frac{1}{K}\left(\gamma_{b,b'}^{+}(S_{b,b'}^{1,1} + S_{b,b'}^{0,0}) - \gamma_{b,b'}^{-}(S_{b,b'}^{1,1} + S_{b,b'}^{0,0} - S_{b,b'}^{\tau}) - S_{b,b'}^{\tau}\right) \\ =&\frac{1}{K}\left((\gamma_{b,b'}^{+} - \gamma_{b,b'}^{-})(S_{b,b'}^{1,1} + S_{b,b'}^{0,0}) - (1-\gamma_{b,b'}^{-})S_{b,b'}^{\tau}\right). \end{aligned} \tag{92}$$

Based on our recursion in Appendix C, we actually have closed-form expressions for the coefficients:

$$\gamma_{b,b'}^{+} - \gamma_{b,b'}^{-} = \begin{cases} 2/\rho, & \delta = 0 \\ 2(2\rho-1)^{\delta-1}, & \delta \geq 1 \end{cases} \tag{93}$$

and

$$1 - \gamma_{b,b'}^{-} = \begin{cases} 1, & \delta = 0 \\ (2\rho-1)^{\delta-1}, & \delta \geq 1 \end{cases} \tag{94}$$

By aggregating all pairs of blocks, we end up with the form in Corollary 2.

F. Proof of Theorem 5

We first show that within each block $b \in [B]$, the average treatment effect $\widehat{\tau}_b$ is asymptotic normal. To see this, we note that there are four groups of possible assignment pairs: $(1,1)$, $(0,0)$, $(1,0)$, $(0,1)$ with fixed number of counts. Let us define the sample means of these four groups by $\widehat{Y}_b(1,1)$, $\widehat{Y}_b(0,0)$, $\widehat{Y}_b(1,0)$, $\widehat{Y}_b(0,1)$. Since these pairs are randomly permuted, following Theorem 2 in Li and Ding (2017), we have

$$\boldsymbol{\theta}_b = \left(\frac{\widehat{Y}_b(1,1) - \bar{Y}_b(1,1)}{\sqrt{\mathrm{Var}(\widehat{Y}_b(1,1))}}, \frac{\widehat{Y}_b(0,0) - \bar{Y}_b(0,0)}{\sqrt{\mathrm{Var}(\widehat{Y}_b(0,0))}}, \frac{\widehat{Y}_b(1,0) - \bar{Y}_b(1,0)}{\sqrt{\mathrm{Var}(\widehat{Y}_b(1,0))}}, \frac{\widehat{Y}_b(0,1)) - \bar{Y}_b(0,1)}{\sqrt{\mathrm{Var}(\widehat{Y}_b(0,1))}} \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{V}). \tag{95}$$

This naturally leads to

$$\frac{\widehat{\tau}_b - \tau_b}{\sqrt{\mathrm{Var}(\widehat{\tau}_b)}} \xrightarrow{d} \mathcal{N}(0,1). \tag{96}$$

Note that our estimator $\widehat{\tau}$ is the average of all blocks:

$$\widehat{\tau} = \frac{1}{B} \sum_{b \in [B]} \widehat{\tau}_b \tag{97}$$

but given that the assignments across blocks are dependent, the asymptotic normality of $\widehat{\tau}$ is not straightforward. In principle, we need to show that $(\widehat{\tau}_b, \forall b \in [B])$ is jointly asymptotic normal. One sufficient condition we leverage here is to show that every pair $(\widehat{\tau}_b, \widehat{\tau}_{b'})$ is jointly normal. We argue this by starting with $b' = b+1$ and it suffices to show that $\boldsymbol{\theta}_{b+1} | \boldsymbol{\theta}_b$ is normal with means that are linear in $\boldsymbol{\theta}_b$. Following our randomization mechanism, conditional on the assignments up to block b, we create two groups where the first group assigns treatment to block b and the second group assigns control to block b. We then conduct a random permutation over $K_v/2$ of 1 and $(K - K_v)/2$ of 0 in the first group of block $b+1$, and another independent random permutation over $K_v/2$ of 0 and $(K - K_v)/2$ of 1 in the second group of block $b+1$. As a result, the conditional averages are still jointly normal. We also note that all the averages are essentially linear operations weighted by the potential outcomes so the means of normal are linear operations as well. Therefore, $(\boldsymbol{\theta}_b, \boldsymbol{\theta}_{b+1})$ are jointly normal. For any pair $b < b' - 1$, we leverage the property of conditional independence:

$$\boldsymbol{\theta}_{b'} \perp \boldsymbol{\theta}_b | \boldsymbol{\theta}_{b'-1} \tag{98}$$

which says that once the block $b'-1$ is realized, the blocks b' and b are independent due to our randomization process. This implies that $\boldsymbol{\theta}_b, \forall b \in [B]$ is a Gaussian Markov Process, and consequently $(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots, \boldsymbol{\theta}_B)$ are jointly normal, so do $(\widehat{\tau}_1, \widehat{\tau}_2, \ldots, \widehat{\tau}_B)$.

G. Proof of Proposition 1

First, we recall that the variance of the estimator (2) satisfies

$$\begin{aligned}
&\mathrm{Var}^{\mathrm{U}}(\widehat{\tau}) = V_1^{\mathrm{U}} + V_2 = \mathrm{Var}(\widehat{\tau}) + \frac{1}{K(K-1)} \sum_{k\in[K]} (\bar{\tau}_k - \bar{\tau})^2 \\
&V_1^{\mathrm{U}} = \frac{1}{B^2 K} \sum_{b,b'\in[B]} (\gamma^+_{b,b'}(S^{1,1}_{b,b'} + S^{0,0}_{b,b'}) - \gamma^-_{b,b'}(S^{1,0}_{b,b'} + S^{0,1}_{b,b'})) \\
&V_2 = \frac{1}{B^2 K^2} \sum_{b,b'\in[B]} \sum_{t\in\mathcal{T}_b, t'\in\mathcal{T}_{b'}} (\gamma^+_{t,t'}(\sigma^{1,1}_{t,t'} + \sigma^{0,0}_{t,t'}) - \gamma^-_{t,t'}(\sigma^{1,0}_{t,t'} + \sigma^{0,1}_{t,t'}))
\end{aligned} \tag{99}$$

where we drop the non-negative variance of treatment effect (which is not identifiable) from (16). Hence, the goal is then to estimate $\mathrm{Var}^{\mathrm{U}}(\widehat{\tau})$.

To estimate the variance, we first consider the following sample statistics for $b = b'$:

$$R_b^1 = \frac{1}{K_v/2 - 1} \sum_{t\in\mathcal{T}_b} \mathbb{1}\{\boldsymbol{w}^{\mathrm{obs}}_{t-1:t} = (1,1)\}(Y_t(\boldsymbol{w}^{\mathrm{obs}}_{1:t-2}, 1, 1) - \widehat{Y_b(\cdot,1,1)})^2 \tag{100}$$

where $\widehat{Y_b(\cdot,1,1)}$ is the sample mean within the block. For ease of analysis, we also define the residual

$$\epsilon_t(\boldsymbol{w}^{\mathrm{obs}}_{1:t-2}, 1, 1) = Y_t(\boldsymbol{w}^{\mathrm{obs}}_{1:t-2}, 1, 1) - \mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2}, 1, 1)]. \tag{101}$$

By taking the expectation, we have

$$\begin{aligned}
\mathbb{E}[R_b^1] =& \frac{1}{K_v/2 - 1} \sum_{t\in\mathcal{T}_b} \mathbb{E}[\mathbb{1}\{\boldsymbol{w}^{\mathrm{obs}}_{t-1:t} = (1,1)\}(Y_t(\boldsymbol{w}^{\mathrm{obs}}_{1:t-2}, 1, 1) - \widehat{Y_b(\cdot,1,1)})^2] \\
=& \frac{1}{K_v/2 - 1} \sum_{t\in\mathcal{T}_b} \mathbb{E}[\mathbb{1}\{\boldsymbol{w}^{\mathrm{obs}}_{t-1:t} = (1,1)\}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2}, 1, 1)] - \mathbb{E}[\widehat{Y_b(\cdot,1,1)}])^2] \\
&+ \frac{1}{K_v/2 - 1} \sum_{t\in\mathcal{T}_b} \mathbb{E}[\mathbb{1}\{\boldsymbol{w}^{\mathrm{obs}}_{t-1:t} = (1,1)\}(\epsilon_t(\boldsymbol{w}^{\mathrm{obs}}_{1:t-2}, 1, 1) - \widehat{\epsilon_b(\cdot,1,1)})^2].
\end{aligned} \tag{102}$$

The first part is equivalent to

$$\begin{aligned}
&\frac{1}{K_v/2 - 1} \sum_{t\in\mathcal{T}_b} \mathbb{E}[\mathbb{1}\{\boldsymbol{w}^{\mathrm{obs}}_{t-1:t} = (1,1)\}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2}, 1, 1)] - \mathbb{E}[\widehat{Y_b(\cdot,1,1)}])^2] \\
=& \frac{1}{K_v/2 - 1} \frac{K_v/2 - 1}{K - 1} \sum_{t\in\mathcal{T}_b} (\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2}, 1, 1)] - \overline{\mathbb{E}[Y_b(\cdot,1,1)]})^2 = S_b^1.
\end{aligned} \tag{103}$$

The second part is equivalent to

$$\begin{aligned}
&\frac{1}{K_v/2-1}\sum_{t\in\mathcal{T}_b}\mathbb{E}[\mathbb{1}\{\boldsymbol{w}_{t-1:t}^{\text{obs}}=(1,1)\}(\epsilon_t(\boldsymbol{w}_{1:t-2}^{\text{obs}},1,1)-\widehat{\epsilon_b(\cdot,1,1)})^2]\\
=&\frac{1}{K_v/2-1}\frac{K_v/2-1}{K}\left(\sum_{t\in\mathcal{T}_b}\text{Var}(\epsilon_t(\boldsymbol{W}_{1:t-2},1,1))-\frac{1}{K-1}\sum_{t'\neq t\in\mathcal{T}_b}\text{Cov}(\epsilon_t(\boldsymbol{W}_{1:t-2},1,1),\epsilon_{t'}(\boldsymbol{W}_{1:t'-2},1,1))\right).\\
=&\frac{1}{K}\sum_{t\in\mathcal{T}_b}\text{Var}(\epsilon_t(\boldsymbol{W}_{1:t-2},1,1))-\frac{1}{K(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}\text{Cov}(\epsilon_t(\boldsymbol{W}_{1:t-2},1,1),\epsilon_{t'}(\boldsymbol{W}_{1:t'-2},1,1))\\
=&\frac{1}{K}\sum_{t\in\mathcal{T}_b}\text{Var}(Y_t(\boldsymbol{W}_{1:t-2},1,1))-\frac{1}{K(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}\text{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))\\
=&\frac{1}{K}\sum_{t\in\mathcal{T}_b}\sigma_t^1-\frac{1}{K(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}\sigma_{t,t'}^{1,1}
\end{aligned}\tag{104}$$

Note that $\gamma_{b,b}^+=\frac{2}{\rho}$ and $\gamma_{t,t'}^+=\frac{2(\rho K/2-1)}{\rho(K-1)}$ for $t'\neq t$, we have

$$\begin{aligned}
\frac{2}{\rho K}\mathbb{E}[R_b^1]=&\frac{2}{\rho K}S_b^1+\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}\sigma_t^1-\frac{2}{\rho K^2(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}\sigma_{t,t'}^{1,1}\\
=&\frac{2}{\rho K}S_b^1+\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}\sigma_t^1+\frac{\rho K-2}{\rho K^2(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}\sigma_{t,t'}^{1,1}-\frac{1}{K(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}\sigma_{t,t'}^{1,1}.
\end{aligned}\tag{105}$$

Recall that the counterpart of variance within a block in our target $\text{Var}^{\text{U}}(\widehat{\tau})$ is

$$\text{Var}^{\text{U}}(\widehat{\tau}_b)=\frac{2}{\rho K}(S_b^1+S_b^0)+\frac{2}{\rho K^2}\sum_{t\in\mathcal{T}_b}(\sigma_t^1+\sigma_t^0)+\frac{1}{K^2}\sum_{t\neq t'\in\mathcal{T}_b}(\gamma_{t,t'}^+(\sigma_{t,t'}^{1,1}+\sigma_{t,t'}^{0,0})-\gamma_{t,t'}^-(\sigma_{t,t'}^{1,0}+\sigma_{t,t'}^{0,1})).\tag{106}$$

Therefore, we have the following decomposition

$$\frac{2}{\rho K}(\mathbb{E}[R_b^1]+\mathbb{E}[R_b^0])=\text{Var}^{\text{U}}(\widehat{\tau}_b)-\frac{1}{K(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}(\sigma_{t,t'}^{1,1}+\sigma_{t,t'}^{0,0}-\sigma_{t,t'}^{1,0}-\sigma_{t,t'}^{0,1}).\tag{107}$$

Based on our analysis of the covariance (73) in Appendix D, we know that

$$\begin{aligned}
-\sum_{t'\neq t\in\mathcal{T}_b}\sigma_{t,t'}^{1,1}=&\frac{\rho(1-\rho)}{K/2-1}\left(\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1)])\right)^2\\
&-\frac{\rho(1-\rho)}{K/2-1}\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1)])^2\\
&-\sum_{t'\neq t\in\mathcal{T}_b}\mathbb{E}[\text{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))|W_{t-2},W_{t'-2}].
\end{aligned}\tag{108}$$

Following the sequential conditioning procedure, we have

$$-\sum_{t'\neq t\in\mathcal{T}_b}\sigma_{t,t'}^{1,1}\geq-\frac{\rho(1-\rho)}{K/2-1}\sum_{q=2}^{\infty}\sum_{t\in\mathcal{T}_b}\phi_t(q)^2.\tag{109}$$

Similarly, we have

$$\sum_{t'\neq t\in\mathcal{T}_b}\sigma^{1,0}_{t,t'}\geq-\frac{\rho(1-\rho)}{K/2-1}\sum_{q=3}^{\infty}\sum_{t\neq t'\in\mathcal{T}_b}\phi_t(q)\phi_{t'}(q). \tag{110}$$

These lead to

$$\frac{2}{\rho K}(\mathbb{E}[R^1_b]+\mathbb{E}[R^0_b])\geq\mathrm{Var}^{\mathrm{U}}(\widehat{\tau}_b)-\frac{2\rho(1-\rho)}{K(K-1)}\left(\frac{K}{K/2-1}\phi(2)^2+\frac{K^2}{K/2-1}\sum_{q=3}^{\infty}\phi(q)^2\right). \tag{111}$$

Then we consider the case of $\delta_{b,b'}\geq 1$ with a similar sample statistic. Let $N^{1,1}_{b,b'}=\sum_{t\in\mathcal{T}_b,t'=t+\delta_{b,b'}}\mathbb{1}\{\boldsymbol{w}^{\mathrm{obs}}_{t-1:t}=\boldsymbol{w}^{\mathrm{obs}}_{t'-1:t'}=(1,1)\}$, and

$$R^{1,1}_{b,b'}=\frac{1}{N^{1,1}_{b,b'}-1}\sum_{t\in\mathcal{T}_b,t'=t+\delta_{b,b'}}\mathbb{1}\{\boldsymbol{w}^{\mathrm{obs}}_{t-1:t}=\boldsymbol{w}^{\mathrm{obs}}_{t'-1:t'}=(1,1)\}(Y_t(\boldsymbol{w}^{\mathrm{obs}}_{1:t-2},1,1)-\widehat{Y_b(\cdot,1,1)})(Y_{t'}(\boldsymbol{w}^{\mathrm{obs}}_{1:t'-2},1,1)-\widehat{Y_{b'}(\cdot,1,1)}). \tag{112}$$

By taking the expectation and applying the same analysis, we have

$$\frac{\gamma^+_{b,b'}}{K}\mathbb{E}[R^{1,1}_{b,b'}]=\frac{\gamma^+_{b,b'}}{K}S^{1,1}_{b,b'}+\sum_{t\in\mathcal{T}_b}\frac{\gamma^+_{b,b'}}{K^2}\sigma^{1,1}_{t,t+\delta_{b,b'}}+\sum_{t\in\mathcal{T}_b,t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\frac{\gamma^+_{t,t'}}{K^2}\sigma^{1,1}_{t,t'}-\sum_{t\in\mathcal{T}_b,t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\frac{1}{K(K-1)}\sigma^{1,1}_{t,t'}. \tag{113}$$

As a result, we can construct the variance estimation

$$\frac{\gamma^+_{b,b'}}{K}(\mathbb{E}[R^{1,1}_{b,b'}]+\mathbb{E}[R^{0,0}_{b,b'}])-\frac{\gamma^-_{b,b'}}{K}(\mathbb{E}[R^{1,0}_{b,b'}]+\mathbb{E}[R^{0,1}_{b,b'}])=\mathrm{Cov}^{\mathrm{U}}(\widehat{\tau}_b,\widehat{\tau}_{b'})-\frac{1}{K(K-1)}\sum_{t'\neq t\in\mathcal{T}_b}(\sigma^{1,1}_{t,t'}+\sigma^{0,0}_{t,t'}-\sigma^{1,0}_{t,t'}-\sigma^{0,1}_{t,t'}). \tag{114}$$

Using the analysis (85) in Appendix D, we have

$$\begin{aligned}
&\frac{\gamma^+_{b,b'}}{K}(\mathbb{E}[R^{1,1}_{b,b'}]+\mathbb{E}[R^{0,0}_{b,b'}])-\frac{\gamma^-_{b,b'}}{K}(\mathbb{E}[R^{1,0}_{b,b'}]+\mathbb{E}[R^{0,1}_{b,b'}])\\
&\geq\mathrm{Cov}^{\mathrm{U}}(\widehat{\tau}_b,\widehat{\tau}_{b'})-\frac{2\rho(1-\rho)}{K(K-1)}\left(\frac{K(K-1)}{K/2-1}\phi(2)\phi(2+\delta_{b,b'})+\frac{K(K-1)}{K/2-1}\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'})\right)\\
&=\mathrm{Cov}^{\mathrm{U}}(\widehat{\tau}_b,\widehat{\tau}_{b'})-\frac{2\rho(1-\rho)}{K/2-1}\left(\phi(2)\phi(2+\delta_{b,b'})+\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'})\right).
\end{aligned} \tag{115}$$

Therefore, we have

$$\begin{aligned}
\mathbb{E}[\widehat{\mathrm{Var}_\eta}(\widehat{\tau})]&=\mathrm{Var}_\eta(\widehat{\tau})+\frac{1}{K(K-1)}\sum_{k=1}^{K}(\bar{\tau}_k-\tau)^2\\
&\quad-\frac{1}{B^2K(K-1)}\sum_{b,b'\in[B]}\sum_{\substack{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}\\|t'-t|\neq\delta_{b,b'}}}(\sigma^{1,1}_{t,t'}+\sigma^{0,0}_{t,t'}-\sigma^{1,0}_{t,t'}-\sigma^{0,1}_{t,t'})\\
&\geq\mathrm{Var}_\eta(\widehat{\tau})-\frac{2\rho(1-\rho)}{B(K-1)(K/2-1)}\left(\phi(2)^2+K\sum_{q=3}^{\infty}\phi(q)^2\right)\\
&\quad-\frac{2\rho(1-\rho)}{B^2(K/2-1)}\sum_{b\neq b'\in[B]}\left(\phi(2)\phi(2+\delta_{b,b'})+\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'})\right).
\end{aligned} \tag{116}$$

H. Proof of Theorem 4

We prove the theorem by adjusting the induction in Appendix A. We first check the block $b=3$. Based on our randomization mechanism, it is guaranteed that we observe $K_v/2$ of $(1,1)$, $K_v/2$ of $(0,0)$, $(K-K_v)/2$ of $(1,0)$, $(K-K_v)/2$ of $(0,1)$. There are $Q=\frac{K!}{(K_v/2)!(K_v/2)!((K-K_v)/2)!((K-K_v)/2)!}$ number of feasible arrangements $\boldsymbol{w}_2^B$ in the block, and it suffices to show that all of them share the same probability $1/Q$. W.L.O.G., let us consider the one

$$\boldsymbol{w}_{2:3}^B=[(1,1),\ldots,(1,0),\ldots,(0,1),\ldots,(0,0),\ldots]. \tag{117}$$

Because the second block is generated as same as the design η, we have $\mathbb{P}(\boldsymbol{W}_2^B=\boldsymbol{w}_2^B)=\frac{(K/2)!(K/2)!}{K!}$. We aim to show that

$$\mathbb{P}(\boldsymbol{W}_{2:3}^B=\boldsymbol{w}_{2:3}^B)=\mathbb{P}(\boldsymbol{W}_2^B=\boldsymbol{w}_2^B)\cdot\mathbb{P}(\boldsymbol{W}_3^B=\boldsymbol{w}_3^B|\boldsymbol{w}_2^B)=1/Q, \tag{118}$$

which requires $\mathbb{P}(\boldsymbol{W}_3^B=\boldsymbol{w}_3^B|\boldsymbol{w}_2^B)=\left[\frac{(K_v/2)!((K-K_v)/2)!}{(K/2)!}\right]^2$. Moreover, it suffices to only check the first half of the block and show that $\mathbb{P}(\boldsymbol{W}_{3,1:K/2}^B=\boldsymbol{w}_{3,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B)=\frac{(K_v/2)!((K-K_v)/2)!}{(K/2)!}$. Instead of showing the exact quantity, we prove it by showing that for two feasible permutations, they are equally likely to be sampled. W.L.O.G., we consider the other one for the first half:

$$\bar{\boldsymbol{w}}_{2:3,1:K/2}^B=[(1,0),\ldots,(1,1)]. \tag{119}$$

Note that $\boldsymbol{w}_{2:3,1:K/2}^B$ and $\bar{\boldsymbol{w}}_{2:3,1:K/2}^B$ share the same assignments for the second block, and their assignments for the third block are reversed. Our goal is to show

$$\mathbb{P}(\boldsymbol{W}_{3,1:K/2}^B=\boldsymbol{w}_{3,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B)=\mathbb{P}(\boldsymbol{W}_{3,1:K/2}^B=\bar{\boldsymbol{w}}_{3,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B)$$

Different from the design η, to generate the third block, now we rely on the assignments of first two blocks, so

$$\mathbb{P}(\boldsymbol{W}_{3,1:K/2}^B=\boldsymbol{w}_{3,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B)=\sum_{\boldsymbol{w}_{1,1:K/2}^B}\mathbb{P}(\boldsymbol{W}_{1,1:K/2}^B=\boldsymbol{w}_{1,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B)\cdot\mathbb{P}(\boldsymbol{W}_{3,1:K/2}^B=\boldsymbol{w}_{3,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B,\boldsymbol{w}_{1,1:K/2}^B). \tag{120}$$

Given that $\boldsymbol{w}_{2,1:K/2}^B=\mathbf{1}$ and the marginal distribution of $\boldsymbol{W}_{1:2}^B$ follows a random permutation, all $\mathbb{P}(\boldsymbol{W}_{1,1:K/2}^B=\boldsymbol{w}_{1,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B)=\frac{(K_v/2)!((K-K_v)/2)!}{(K/2)!}$ share the same conditional probability. Therefore, it suffices to show that

$$\sum_{\boldsymbol{w}_{1,1:K/2}^B}\mathbb{P}(\boldsymbol{W}_{3,1:K/2}^B=\boldsymbol{w}_{3,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B,\boldsymbol{w}_{1,1:K/2}^B)=\sum_{\boldsymbol{w}_{1,1:K/2}^B}\mathbb{P}(\boldsymbol{W}_{3,1:K/2}^B=\bar{\boldsymbol{w}}_{3,1:K/2}^B|\boldsymbol{w}_{2,1:K/2}^B,\boldsymbol{w}_{1,1:K/2}^B). \tag{121}$$

For any $\boldsymbol{w}^B_{1,1:K/2}$, we want to pair it with another $\bar{\boldsymbol{w}}^B_{1,1:K/2}$ so that their conditional probabilities coincide. We make it by construction:

$$\bar{w}^B_{1,k} = \begin{cases} w^B_{1,k}, & w^B_{3,k} = \bar{w}^B_{3,k} \\ 1 - w^B_{1,k}, & w^B_{3,k} \neq \bar{w}^B_{3,k} \end{cases}, \forall k \in [K/2]. \tag{122}$$

In this way, we have

$$\mathbb{P}(\boldsymbol{W}^B_{3,1:K/2} = \boldsymbol{w}^B_{3,1:K/2} | \boldsymbol{w}^B_{2,1:K/2}, \boldsymbol{w}^B_{1,1:K/2}) = \mathbb{P}(\boldsymbol{W}^B_{3,1:K/2} = \bar{\boldsymbol{w}}^B_{3,1:K/2} | \boldsymbol{w}^B_{2,1:K/2}, \bar{\boldsymbol{w}}^B_{1,1:K/2}) \tag{123}$$

and (122) is true.

Next, suppose that the property is true for some $b \geq 3$, we check the block $b+1$. Because we have a random permutation over $K_v/2$ of $(1,1)$, $K_v/2$ of $(0,0)$, $(K-K_v)/2$ of $(1,0)$, $(K-K_v)/2$ of $(0,1)$ for the block b, it directly implies that the assignments $\boldsymbol{W}^B_b$ is a random permutation over $K/2$ of 1 and $K/2$ of 0. Then following the same argument above, we can show that $\boldsymbol{W}^B_{b:b+1}$ satisfies the property.

I. Proof of Corollary 4

The framework of variance decomposition is exactly the same as Appendix C. Specifically, we have the first part of the variance

$$V_1 = \frac{1}{B^2 K} \sum_{b,b' \in [B]} (\gamma^+_{b,b'}(\rho, \rho^+)(S^{1,1}_{b,b'} + S^{0,0}_{b,b'}) - \gamma^-_{b,b'}(\rho, \rho^+)(S^{1,0}_{b,b'} + S^{0,1}_{b,b'})) - \frac{1}{K(K-1)} \sum_{k=1}^{K} (\bar{\tau}_k - \bar{\tau})^2. \tag{124}$$

The key is to characterize $\gamma^+_{b,b'}(\rho, \rho^+)$ and $\gamma^-_{b,b'}(\rho, \rho^+)$. To do this, we adjust the recursion in Appendix C and obtain

$$\begin{aligned} q_{11,11}(\delta) =& q_{11,11}(\delta-1)\rho^- + q_{11,01}(\delta-1)\rho^+ \\ q_{11,00}(\delta) =& q_{11,00}(\delta-1)\rho^- + q_{11,10}(\delta-1)\rho^+ \\ q_{11,01}(\delta) =& q_{11,00}(\delta-1)(1-\rho^-) + q_{11,10}(\delta-1)(1-\rho^+) \\ q_{11,10}(\delta) =& q_{11,11}(\delta-1)(1-\rho^-) + q_{11,01}(\delta-1)(1-\rho^+) \end{aligned} \tag{125}$$

which has an equivalent matrix representation:

$$\boldsymbol{Q}(\delta) = (q_{11,00}(\delta), q_{11,01}(\delta)), q_{11,10}(\delta), q_{11,11}(\delta))' = \begin{pmatrix} \rho^-, & 0, & \rho^+, & 0 \\ 1-\rho^-, & 0, & 1-\rho^+, & 0 \\ 0, & 1-\rho^+, & 0, & 1-\rho^- \\ 0, & \rho^+, & 0, & \rho^- \end{pmatrix} \cdot \boldsymbol{Q}(\delta-1). \tag{126}$$

If we let Θ denote the matrix with those ρ^- and ρ^+, then we have

$$\boldsymbol{Q}(\delta) = \Theta^{\delta-1} \cdot \left(0, 0, \frac{\rho(1-\rho^-)}{2}, \frac{\rho\rho^-}{2}\right)', \tag{127}$$

which gives us $\gamma^+_{b,b'}(\rho,\rho^+)=4q_{11,11}(\delta_{b,b'})/\rho^2$ and $\gamma^-_{b,b'}(\rho,\rho^+)=4q_{11,00}(\delta_{b,b'})/\rho^2$. Under Assumption 3, the first part of the variance can be reformulated as

$$V_1=\frac{1}{B^2K}\sum_{b,b'\in[B]}((\gamma^+_{b,b'}(\rho,\rho^+)-\gamma^-_{b,b'}(\rho,\rho^+))(S^{1,1}_{b,b'}+S^{0,0}_{b,b'})-(1-\gamma^-_{b,b'}(\rho,\rho^+))S^{\tau}_{b,b'}. \tag{128}$$

Next, we have the second part of the variance

$$V_2=\frac{1}{B^2K^2}\sum_{b,b'\in[B]}\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}(\gamma^+_{t,t'}(\rho,\rho^+)(\sigma^{1,1}_{t,t'}+\sigma^{0,0}_{t,t'})-\gamma^-_{t,t'}(\rho,\rho^+)(\sigma^{1,0}_{t,t'}+\sigma^{0,1}_{t,t'})). \tag{129}$$

To derive the upper bound, we still start with $b=b'$. Following the analysis in Appendix D, we have

$$\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-2},1,1))=\mathrm{Var}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)|W_{t-2}])+\rho^-\,\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-3},1,1,1))+(1-\rho^-)\,\mathrm{Var}(Y_t(\boldsymbol{W}_{1:t-3},0,1,1)). \tag{130}$$

and

$$\mathrm{Var}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)|W_{t-2}])=\rho^-(1-\rho^-)(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1))])^2. \tag{131}$$

where ρ is replaced with ρ^- in the refined design because of how we randomize the assignments. Applying a similar adjustment for $\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))$, we have

$$\begin{aligned}
&\frac{1}{K^2}\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}\gamma^+_{t,t'}(\rho,\rho^+)\,\mathrm{Cov}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-2},1,1)|W_{t-2},W_{t'-2}],\mathbb{E}[Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1)|W_{t-2},W_{t'-2}])\\
\le&\frac{2}{\rho K^2}\left(\rho^-(1-\rho^-)+\frac{\rho K/2-1}{K-1}\frac{\rho^-(1-\rho^-)}{\rho K/2-1}\right)\sum_{t\in\mathcal{T}_b}(\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},1,1,1)]-\mathbb{E}[Y_t(\boldsymbol{W}_{1:t-3},0,1,1)])^2\\
\le&\frac{2}{\rho K^2}\frac{K\rho^-(1-\rho^-)}{K-1}\sum_{t\in\mathcal{T}_b}\phi_t(2)^2\le\frac{2\rho^-(1-\rho^-)}{\rho K(K-1)}\sum_{t\in\mathcal{T}_b}\phi_t(2)^2
\end{aligned} \tag{132}$$

This further leads to

$$\begin{aligned}
\frac{1}{K^2}\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}\gamma^+_{t,t'}(\rho,\rho^+)\,\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},1,1))\le&\frac{2\rho^-(1-\rho^-)}{\rho K(K-1)}\sum_{t\in\mathcal{T}_b}\sum_{q=2}^{\infty}\phi_t(q)^2\\
\le&\frac{2\rho^-(1-\rho^-)}{\rho(K-1)}\sum_{q=2}^{\infty}\phi(q)^2
\end{aligned} \tag{133}$$

Following the same procedure, we also have

$$\begin{aligned}
-\frac{2}{K^2}\sum_{t\in\mathcal{T}_b,t'\in\mathcal{T}_{b'}}\gamma^-_{t,t'}(\rho,\rho^+)\,\mathrm{Cov}(Y_t(\boldsymbol{W}_{1:t-2},1,1),Y_{t'}(\boldsymbol{W}_{1:t'-2},0,0))\le&\frac{2}{K(K-1)}\frac{\rho^-(1-\rho^-)}{\rho K/2-1}\sum_{t\neq t'\in\mathcal{T}_b}\sum_{q=3}^{\infty}\phi_t(q)\phi_{t'}(q)\\
\le&\frac{2\rho^-(1-\rho^-)}{\rho K/2-1}\sum_{q=3}^{\infty}\phi(q)^2.
\end{aligned} \tag{134}$$

Generalizing this to $b \neq b'$, we obtain

$$\begin{aligned}
&\frac{2(\gamma^+_{b,b'}(\rho,\rho^+)+\gamma^-_{b,b'}(\rho,\rho^+))\rho^-(1-\rho^-)}{K^2}\sum_{q=2}^{\infty}\sum_{t\in\mathcal{T}_b}\phi_t(q)\phi_{t+\delta_{b,b'}}(q+\delta_{b,b'})\\
&\quad+\frac{2\rho^-(1-\rho^-)}{K(K-1)(\rho K/2-1)}\sum_{t\in\mathcal{T}_b}\sum_{t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\phi_t(2)\phi_{t'}(2+\delta_{b,b'})\\
&+\frac{2(\gamma^+_{b,b'}(\rho,\rho^+)+\gamma^-_{b,b'}(\rho,\rho^+))\rho^-(1-\rho^-)}{K^2(\rho K/2-1)}\sum_{q=3}^{\infty}\sum_{t\in\mathcal{T}_b}\sum_{t'\neq t+\delta_{b,b'}\in\mathcal{T}_{b'}}\phi_t(q)\phi_{t'}(q+\delta_{b,b'}).
\end{aligned} \tag{135}$$

Since $\gamma^+_{b,b'}(\rho,\rho^+)+\gamma^-_{b,b'}(\rho,\rho^+)=2$, it is further bounded by

$$\frac{4\rho^-(1-\rho^-)}{K}\sum_{q=2}^{\infty}\phi(q)\phi(q+\delta_{b,b'})+\frac{2\rho^-(1-\rho^-)}{\rho K/2-1}\phi(2)\phi(2+\delta_{b,b'})+\frac{4\rho^-(1-\rho^-)}{\rho K/2-1}\sum_{q=3}^{\infty}\phi(q)\phi(q+\delta_{b,b'}). \tag{136}$$

Finally, we end up with the upper bound

$$\begin{aligned}
V_2^{\mathrm{U}} =&\frac{4\rho^-(1-\rho^-)}{B\rho(K-1)}\phi(2)^2+\frac{4\rho^-(1-\rho^-)}{B(\rho K/2-1)}\sum_{q=3}^{\infty}\phi(q)^2\\
&+\frac{4\rho^-(1-\rho^-)}{B^2}\left(\frac{1}{K}+\frac{1}{\rho K-2}\right)\sum_{b\neq b'\in[B]}\phi(2)\phi(2+\delta_{b,b'})+\frac{4\rho^-(1-\rho^-)}{B^2}\left(\frac{1}{K}+\frac{1}{\rho K/2-1}\right)\sum_{b\neq b'\in[B]}\sum_{q=3}^{\infty}\phi(q)\phi(q+
\end{aligned} \tag{137}$$